

PERMANENT MAXILLARY SECOND PREMOLAR

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Maxillary Second Premolar

Morphology and Identification | Dental Anatomy | First-Year BDS

ANTERIOR TO
MOLAR FUNCTION

A key member
of the premolar
family

*Maxillary
Second
Premolar*



Buccal



Lingual



Mesial



Distal



Occlusal



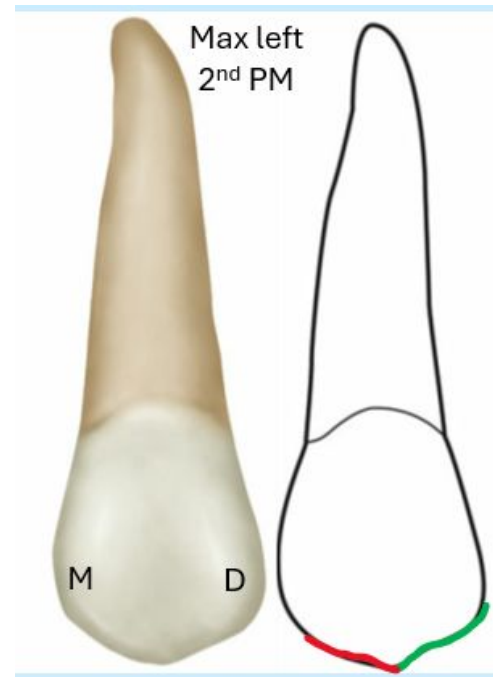
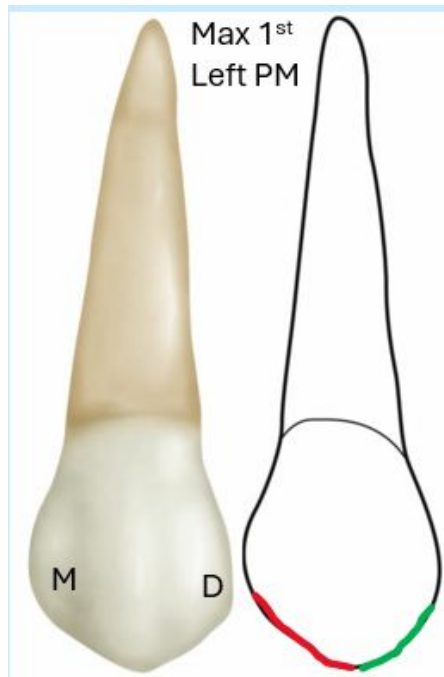
Focus of today

- General morphology
- All aspects
- Comparison with first premolar
- Clinical and exam relevance

*Form
Guides
Function*

Learning Objectives: By the end of this lecture, you should be able to:

1. Identify the maxillary second premolar
2. Describe crown and root morphology
3. Explain buccal, lingual, mesial, distal, and occlusal aspects
4. Distinguish it from the maxillary first premolar
5. Recall chronology, dimensions, and common variations
6. Apply features in viva and practical identification



How would you distinguish a maxillary second premolar from a maxillary first premolar?

Introduction

The maxillary second premolar resembles the maxillary first premolar and complements it in function.

It has a more rounded form and commonly has a single root.

Supplements the maxillary first premolar in function

Resembles the first premolar but is less angular and more rounded

Usually has a single root



First evidence
of calcification



2–2¼ years



Enamel
completed



6–7 years



Eruption



10–12 years



Root
completed



12–14 years

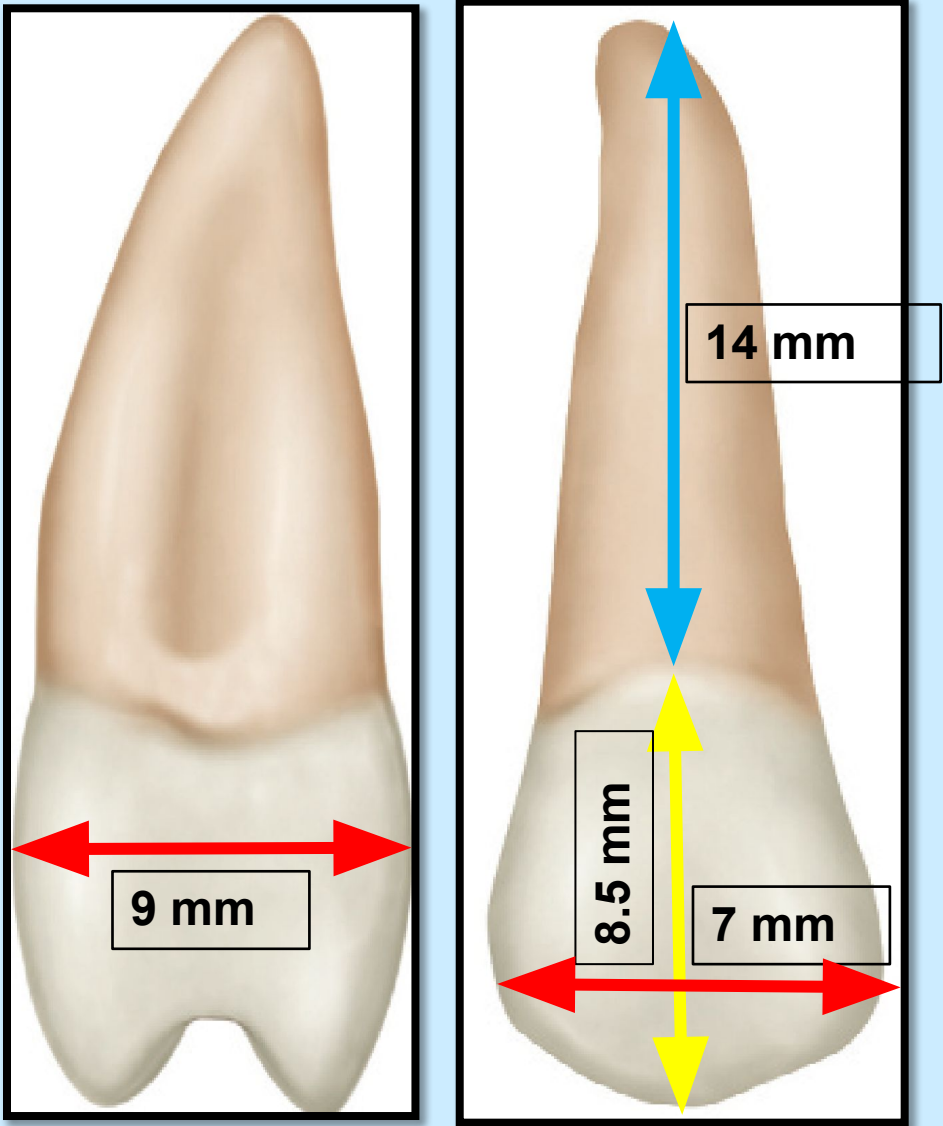
Development Timeline

CHRONOLOGY OF THE PERMANENT SUCCEDANEOUS DENTITION

		1st Evidence Of Calcification	Crown Completed (Years)	Eruption (years)	Root Completed (Years)
Central Incisor	Max.	3-4 months	4-5	7-8	10
1st Premolar	Max.	1½–1¾ years	5-6	10-11	12-13
	Mand.	1¼–2 years	5-6	10-12	12-13
2nd Premolar	Max.	2–2¼ years	6-7	10-12	12-14
			4-5 yrs		

- Both teeth have similar average dimensions
- Root length ~14.0 mm, sometimes exceeding that of 1st PM

Dimensions	Max. 2 nd PM	Max1st PM
Crown length	8.5 mm	8.5 mm
Root length	~14 mm	14 mm
MD Diameter of Crown	7 mm	7 mm
LL Diameter of crown	9 mm	9 mm



General Morphology

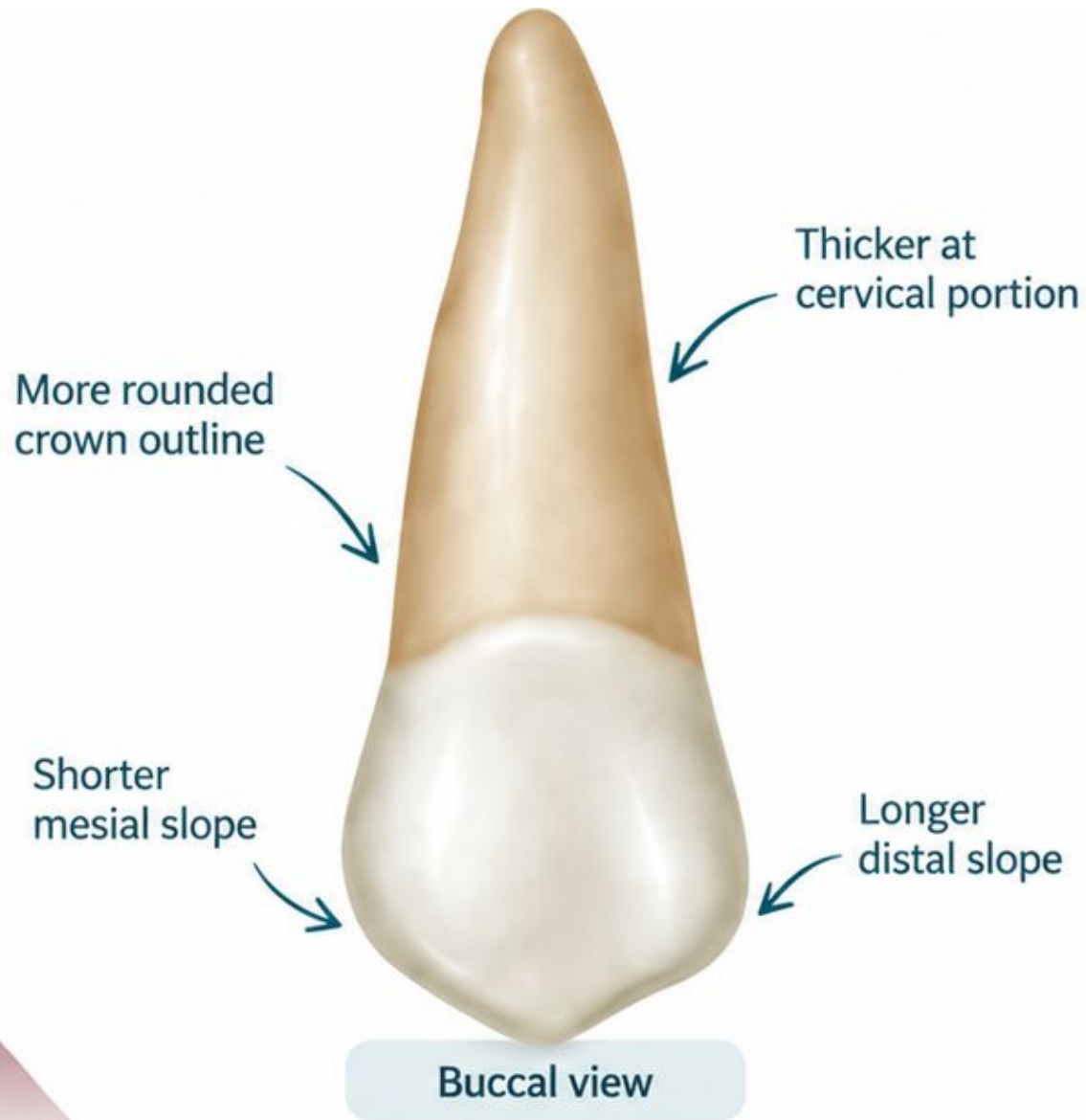
Crown is less angular and more rounded than the maxillary first premolar

Usually has a single tapered root

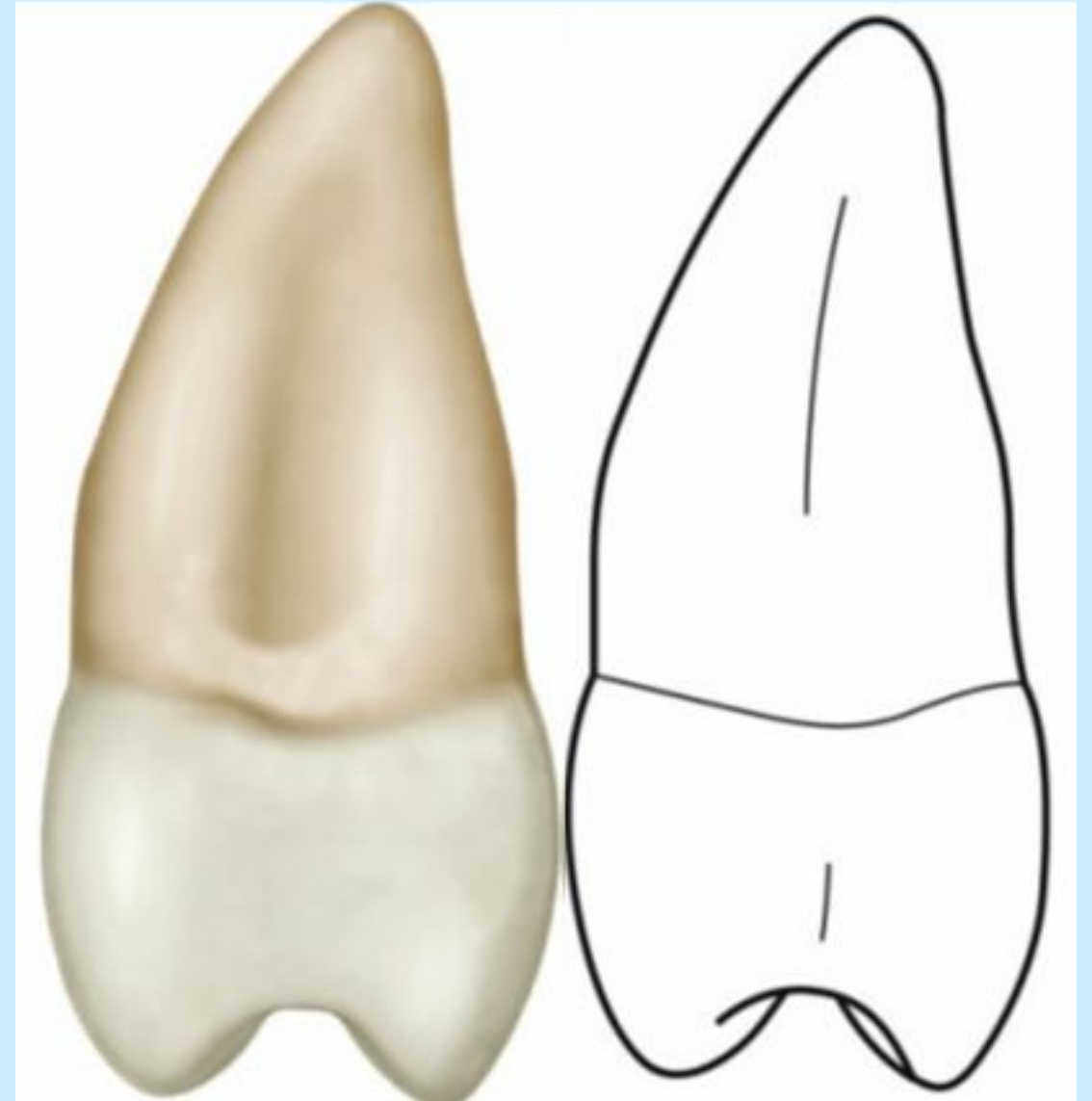
Buccal and lingual cusps are more nearly equal in height

Crown and root may appear relatively thick at the cervical portion

Root is about as long as, or slightly longer than, that of the first premolar

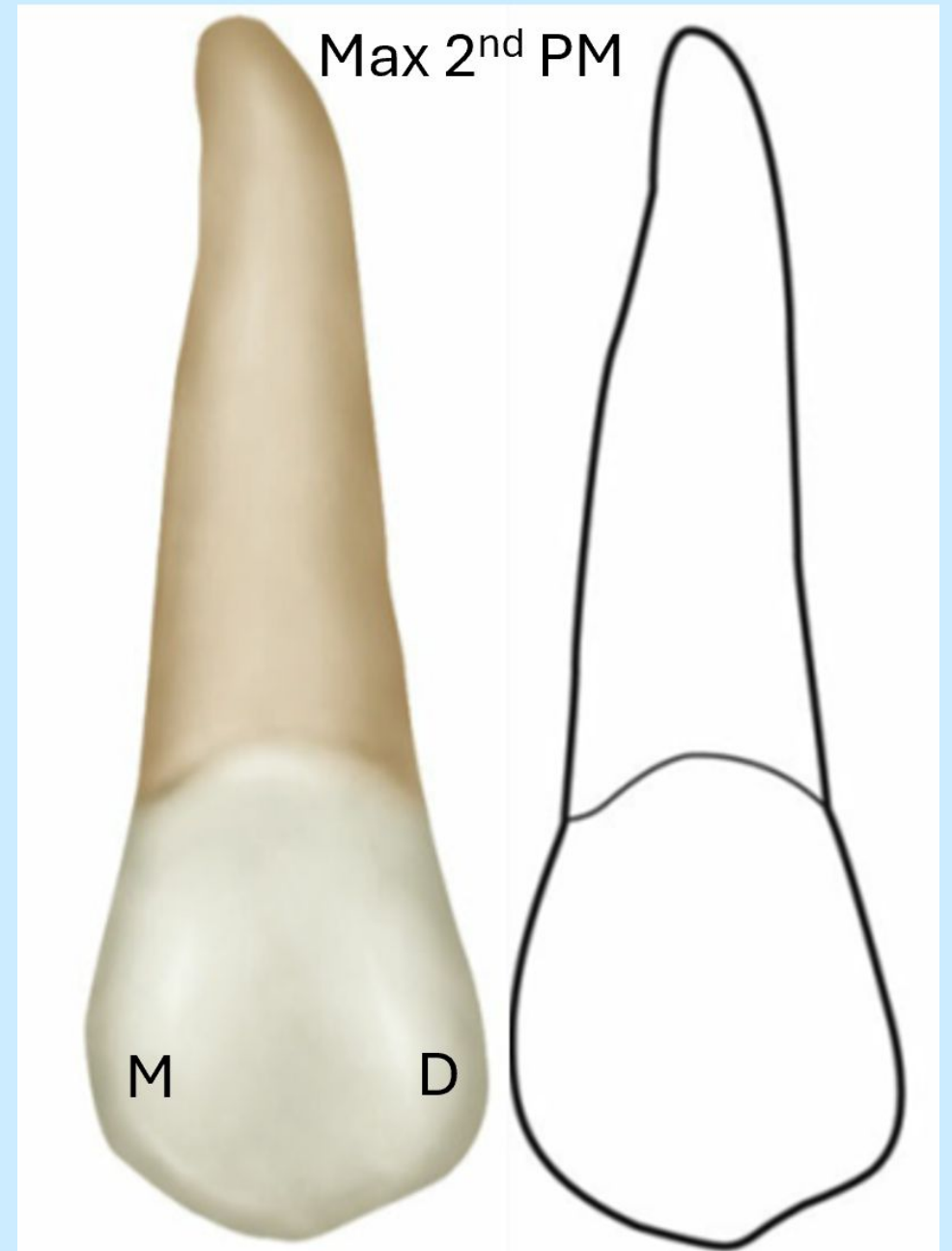


- Two cusps (facial & lingual) of **nearly equal size and height**
- Supplements the 1st premolar in function



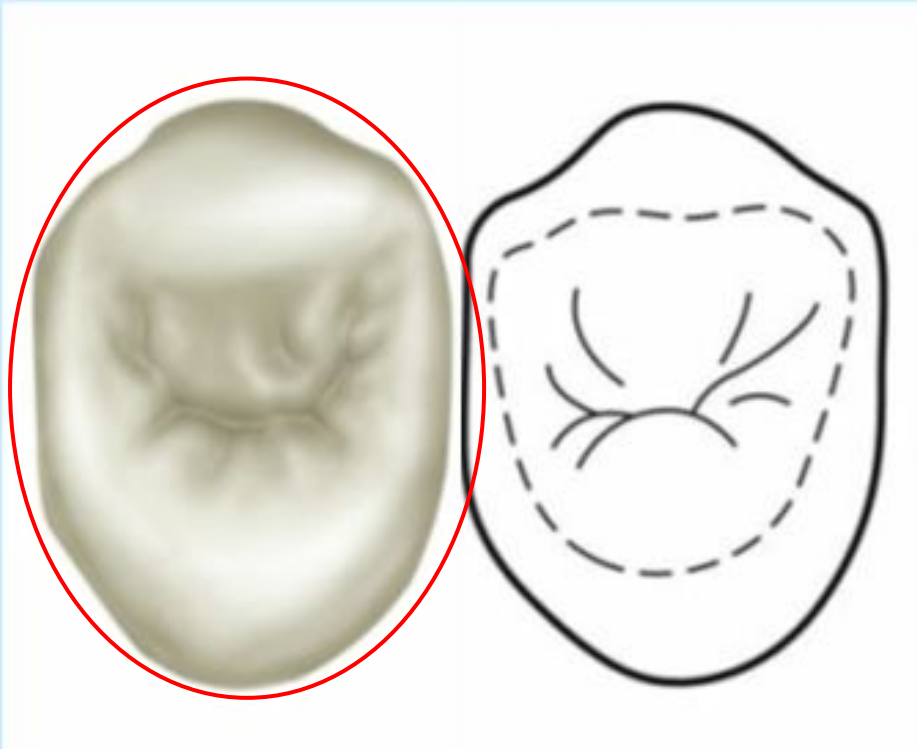
Maxillary left second premolar

- Greater variability than the 1st premolar
- **Distal slope of distal cusp ridge longer than mesial slope of mesial cusp ridge**

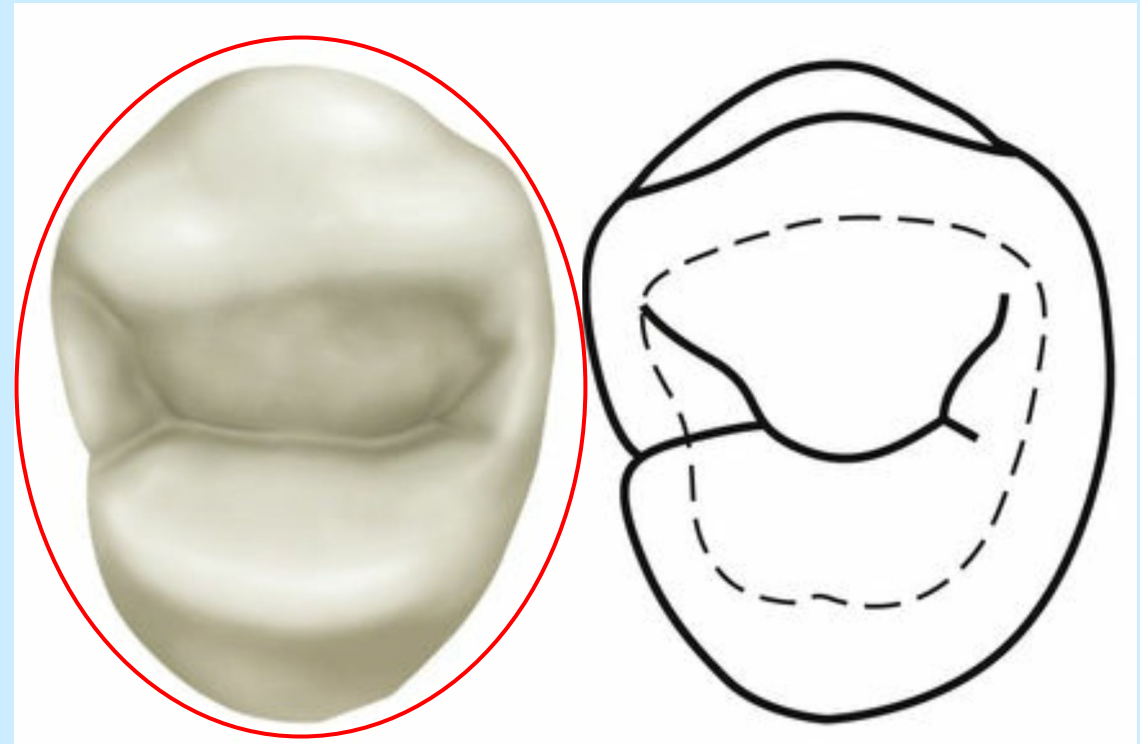


Maxillary left second premolar

- Greater variability than the 1st premolar
- Crown outline and mesiodistal dimensions may be **smaller or larger**



Maxillary left second premolar

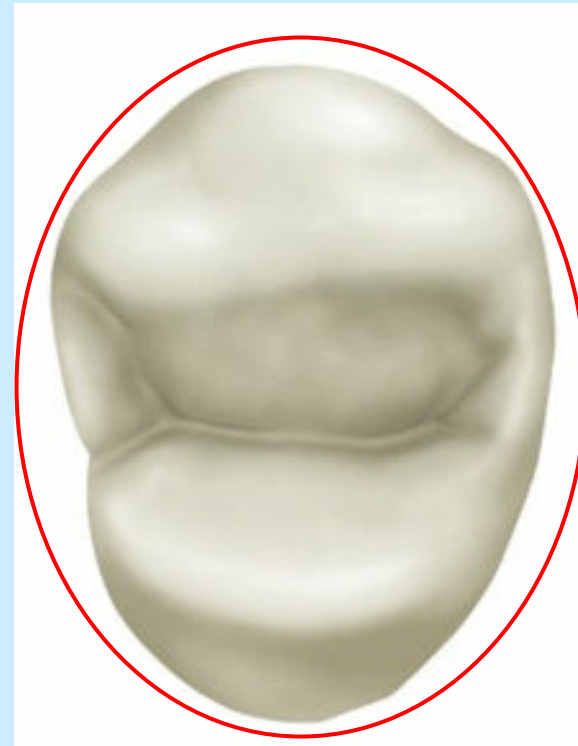


Maxillary left First premolar

- Greater variability than the 1st premolar
- Less angular, more rounded morphology

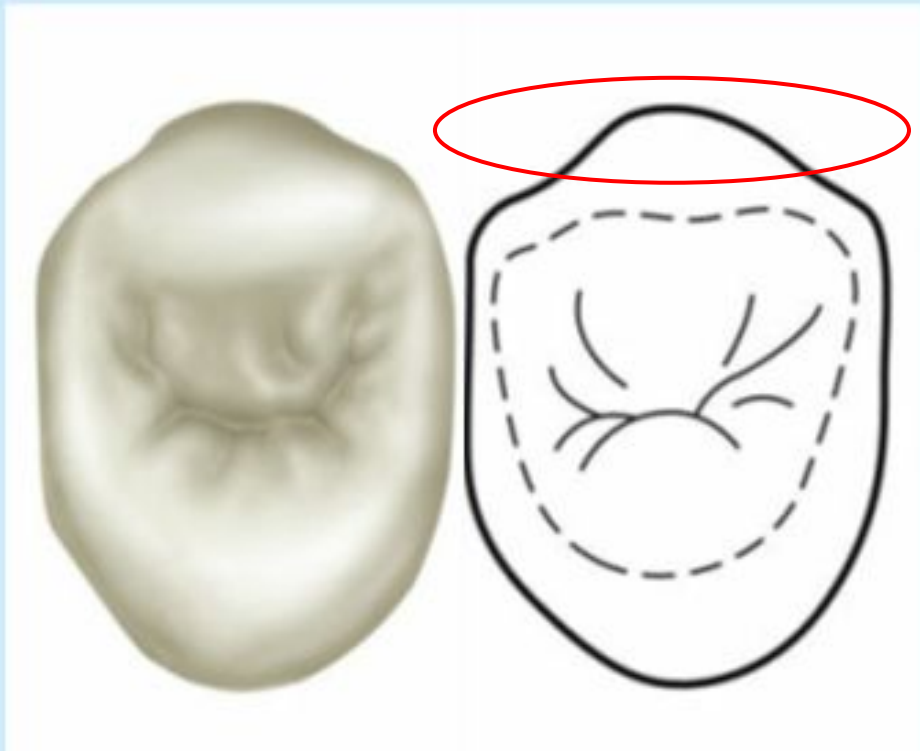


Maxillary left second premolar

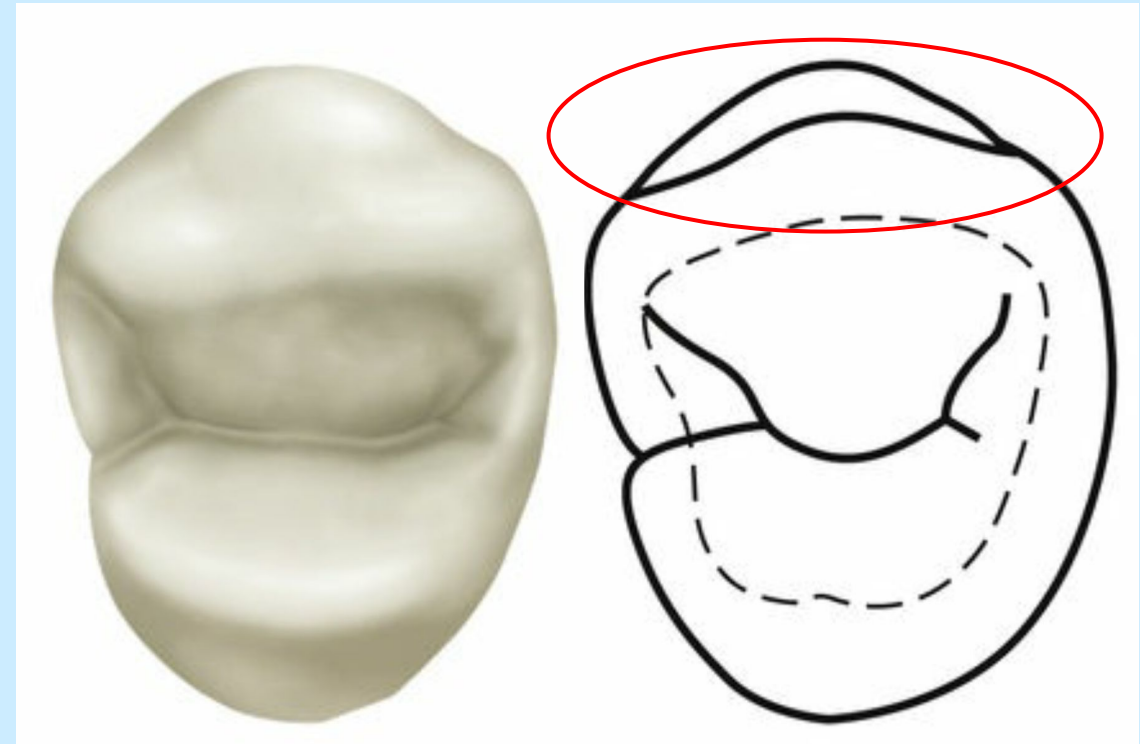


Maxillary left First premolar

- Greater variability than the 1st premolar
- **Facial ridge less prominent than in the 1st premolar**



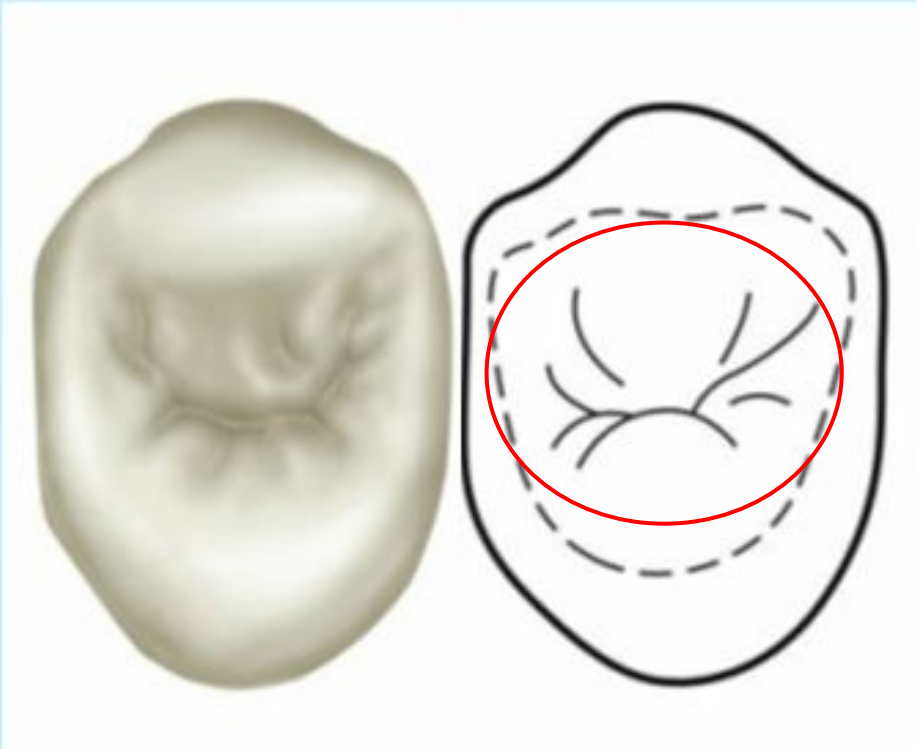
Maxillary left second premolar



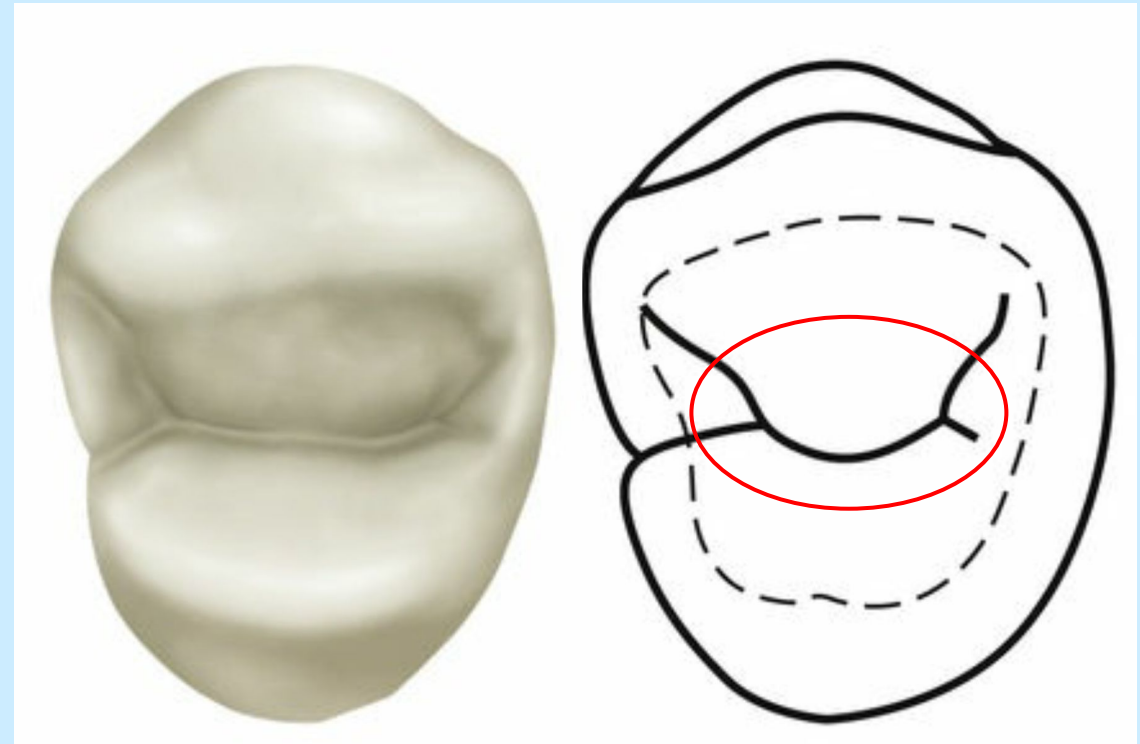
Maxillary left First premolar

Compared to 1st PM, 2nd PM shows:

- More supplemental occlusal anatomy
- Additional secondary ridges and grooves



Maxillary left second premolar

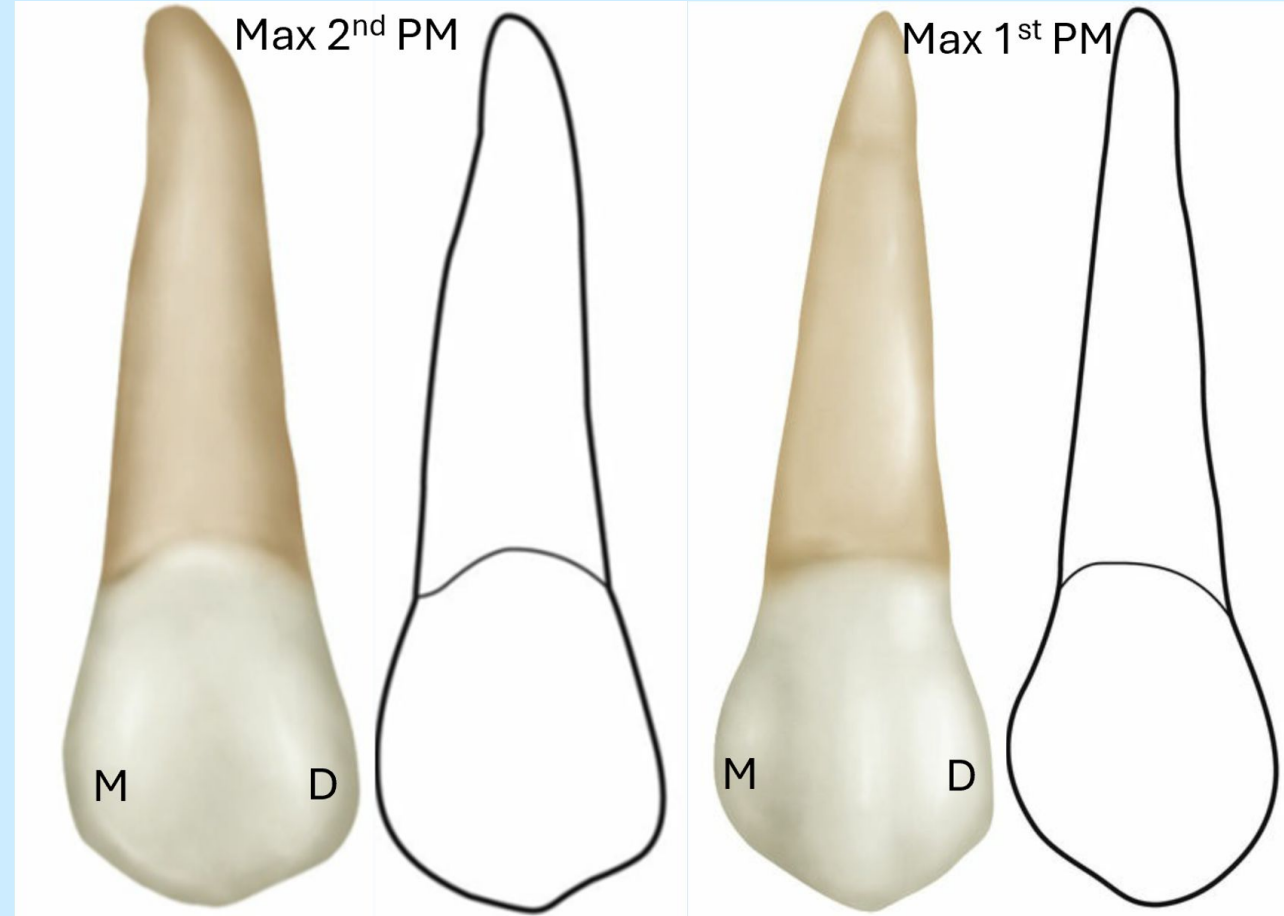


Maxillary left First premolar



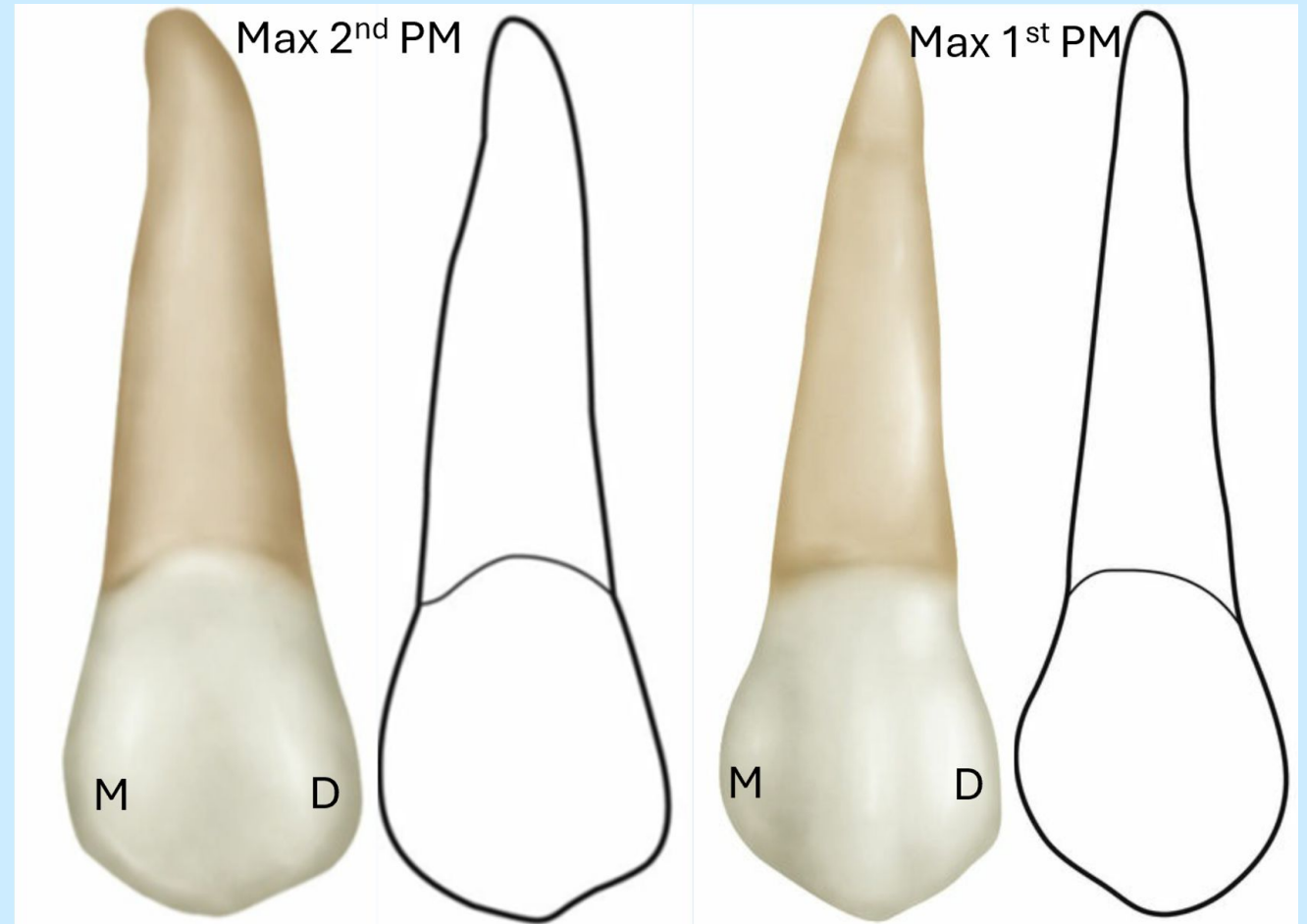
Buccal Aspect

□ Crown outline
appears more
rounded



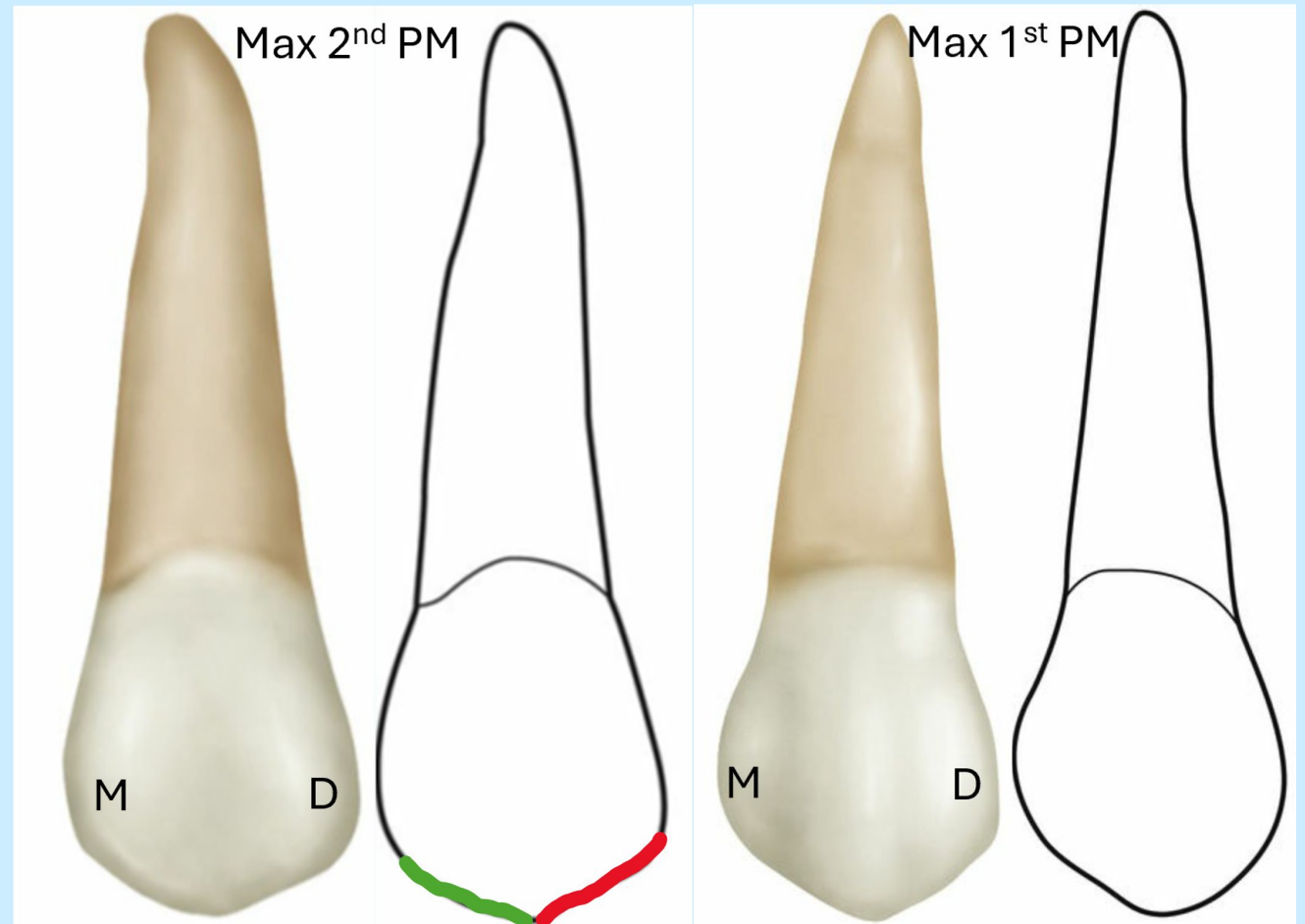
Buccal Aspect

- Buccal cusp is shorter and less pointed than in the maxillary first premolar



Buccal Aspect

- Mesial cusp slope is usually shorter than the distal cusp slope
- that is opposite of max. 1st PM.

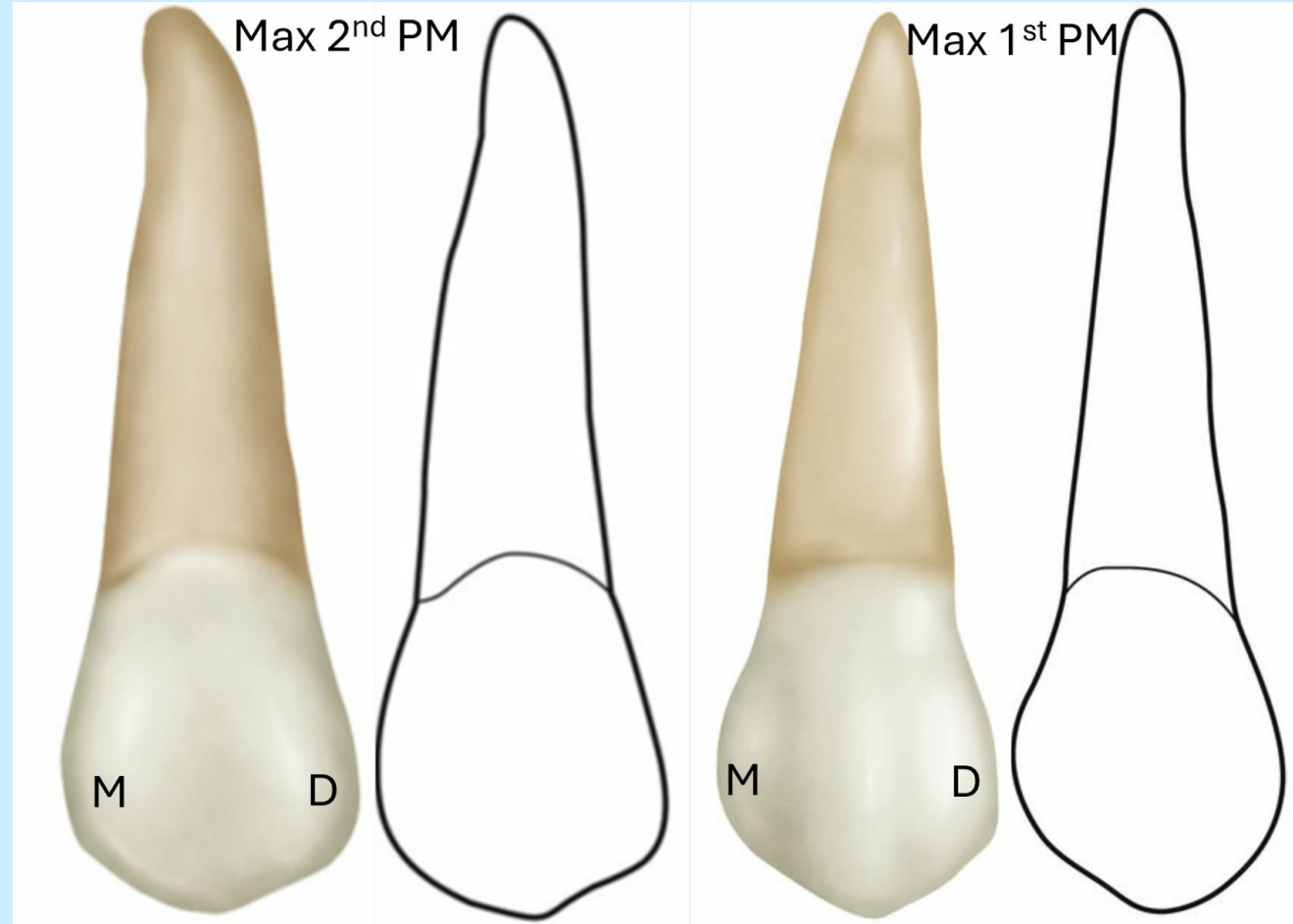


Key identification clue

Short mesial cusp slope +
longer distal cusp slope

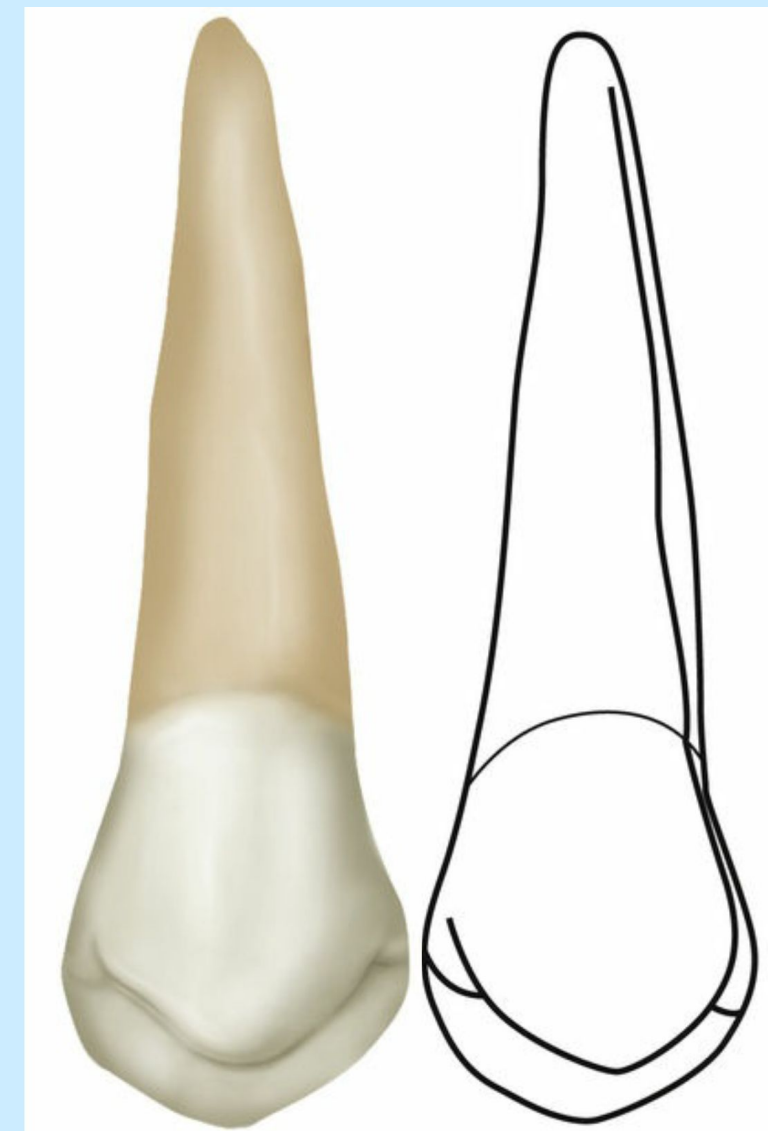
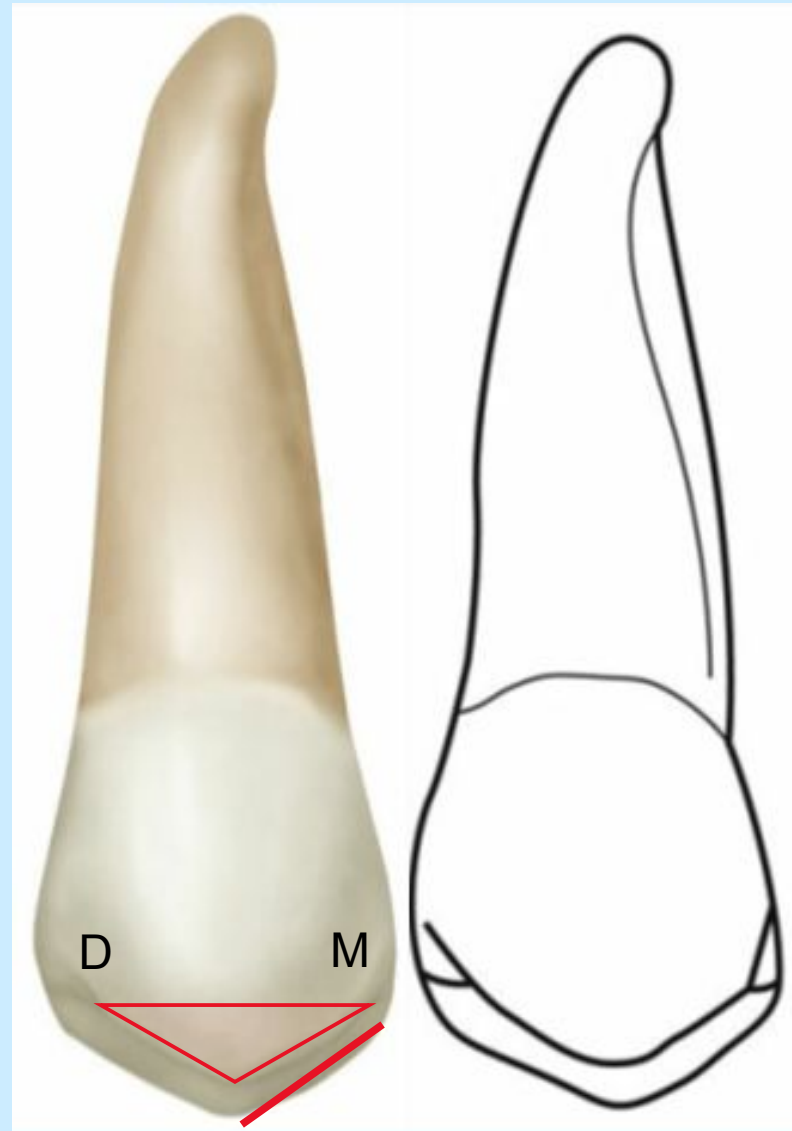
Buccal Aspect

- Buccal ridge is usually less prominent
- Cervical region: Crown and root may appear thicker at the cervical portion (occasionally)



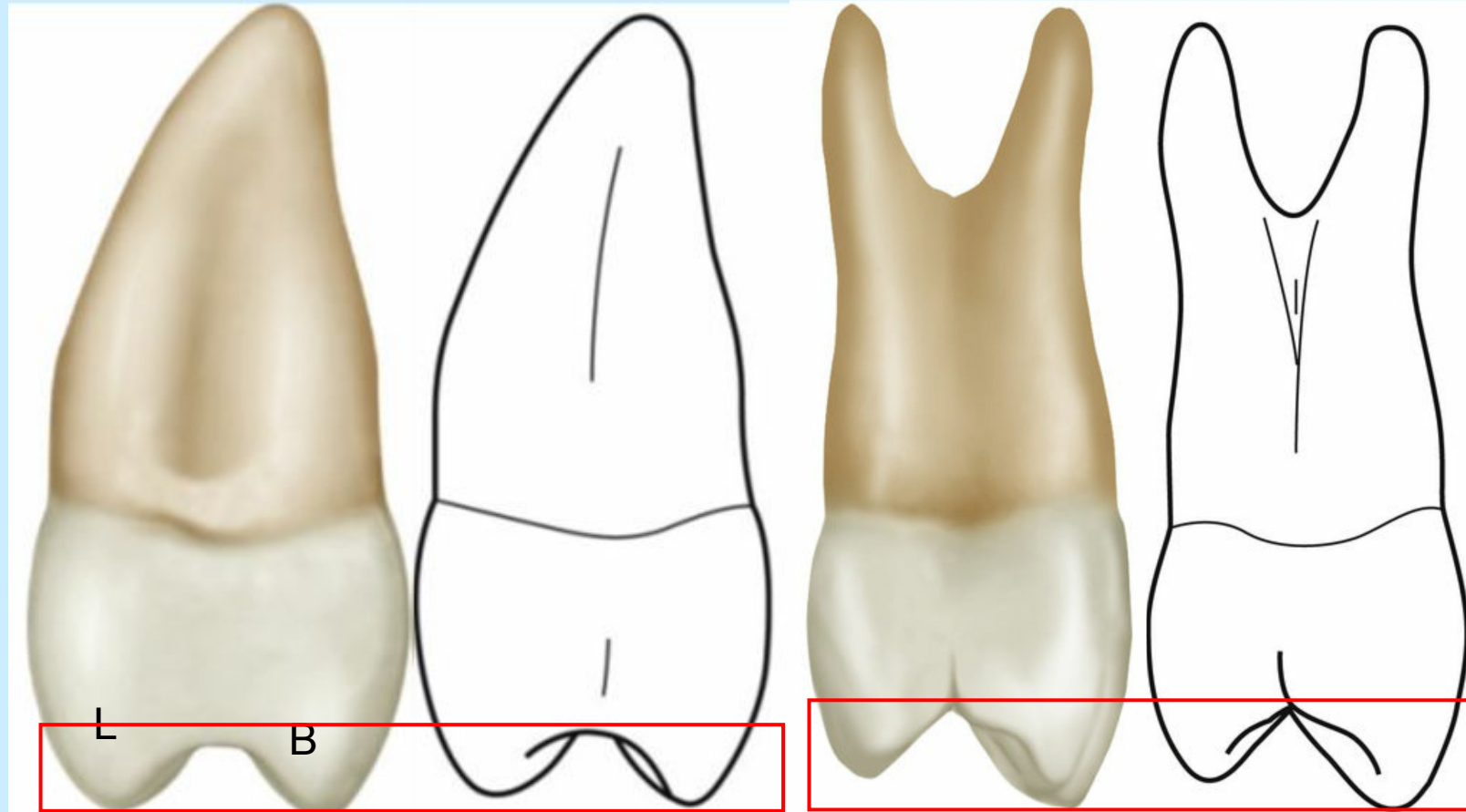
Lingual Aspect

- Lingual cusp is relatively well developed and appears longer than in the first premolar
- Which makes the crown longer on the L side



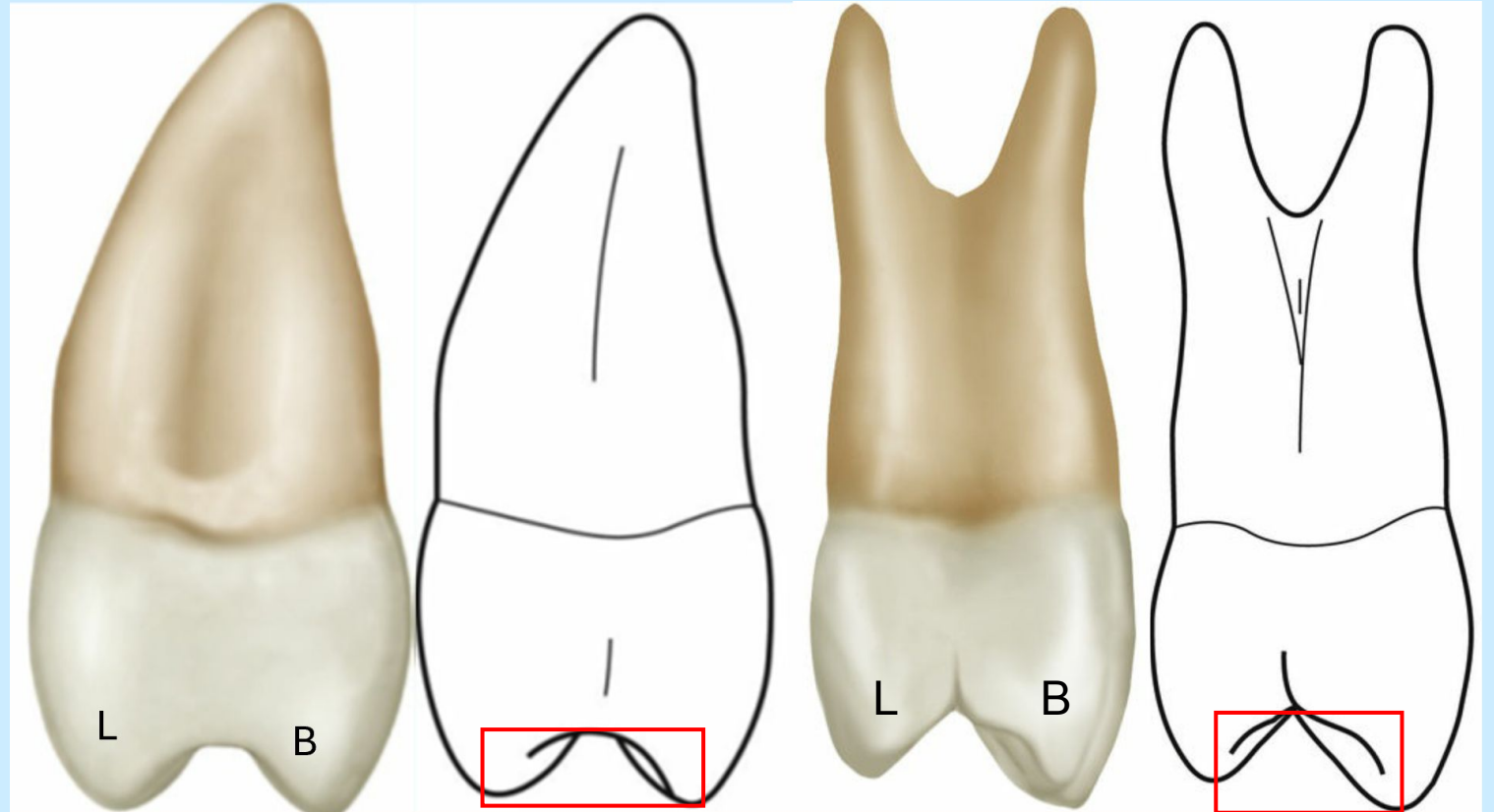
Mesial Aspect

- Cuspal height shorter than max 1st premolar
- Buccal and lingual cusps are more nearly the same length



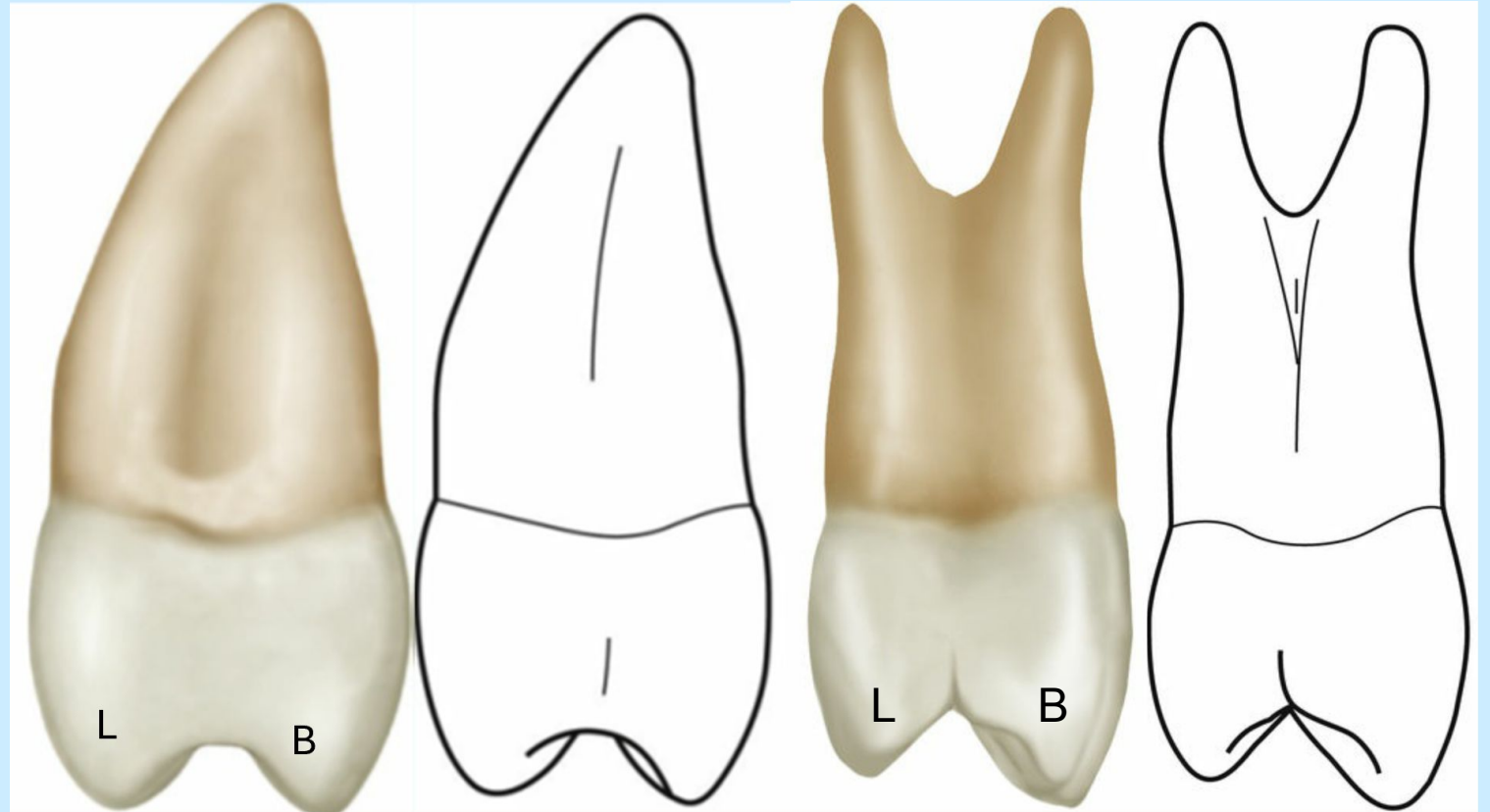
Mesial Aspect

- Greater distance between cusp tips widens the occlusal surface buccolingually



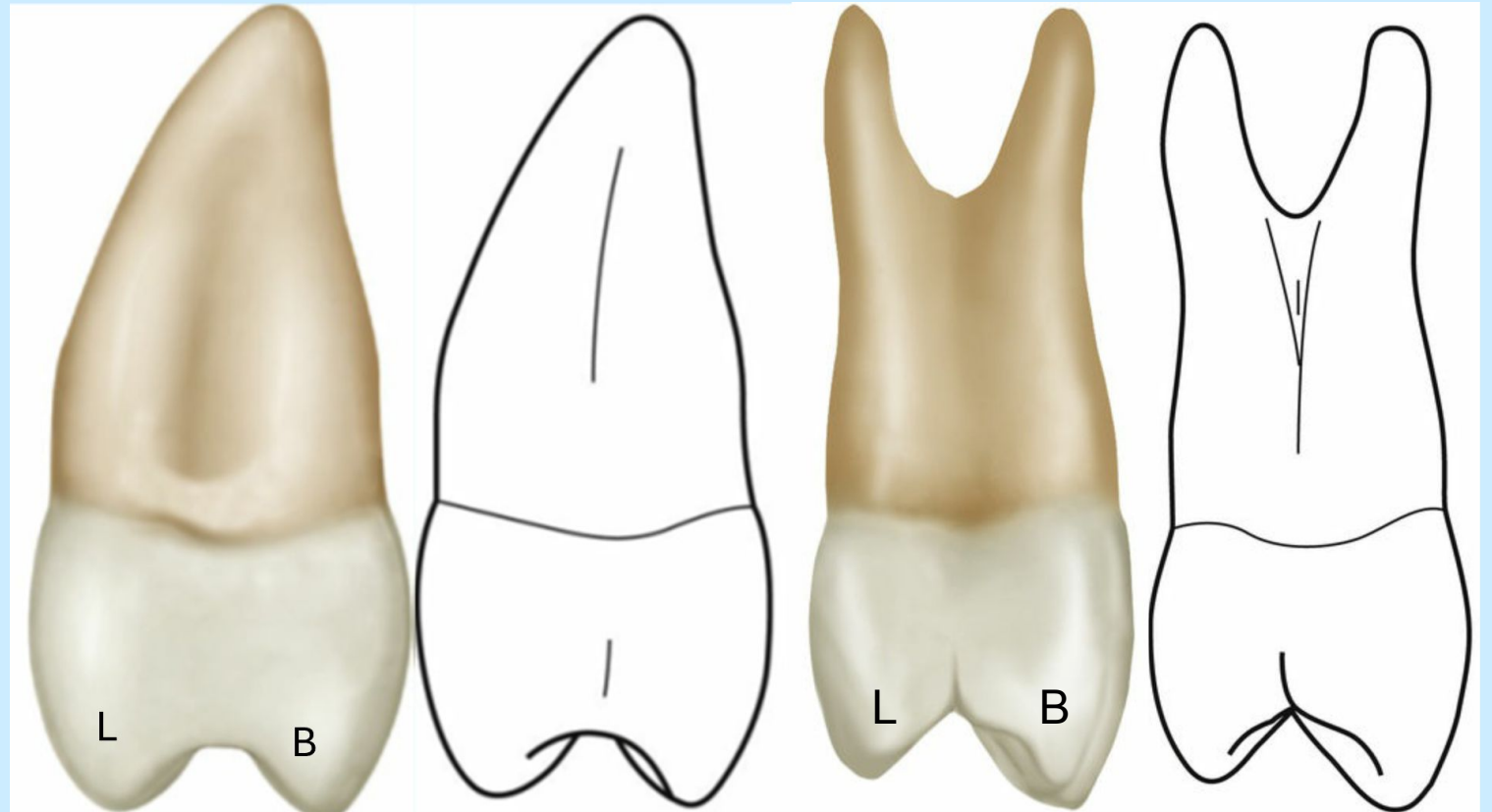
Mesial Aspect

- No mesial marginal developmental groove present



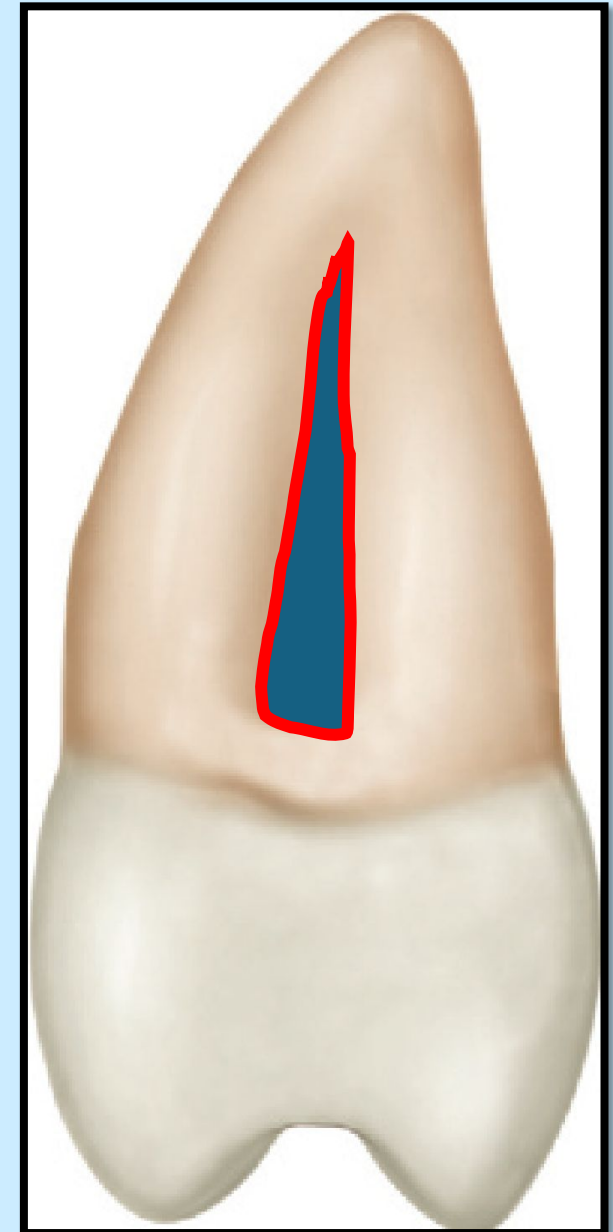
Mesial Aspect

- Crown convex at all points; no developmental depression
- Mesial Root surface shows a shallow developmental groove



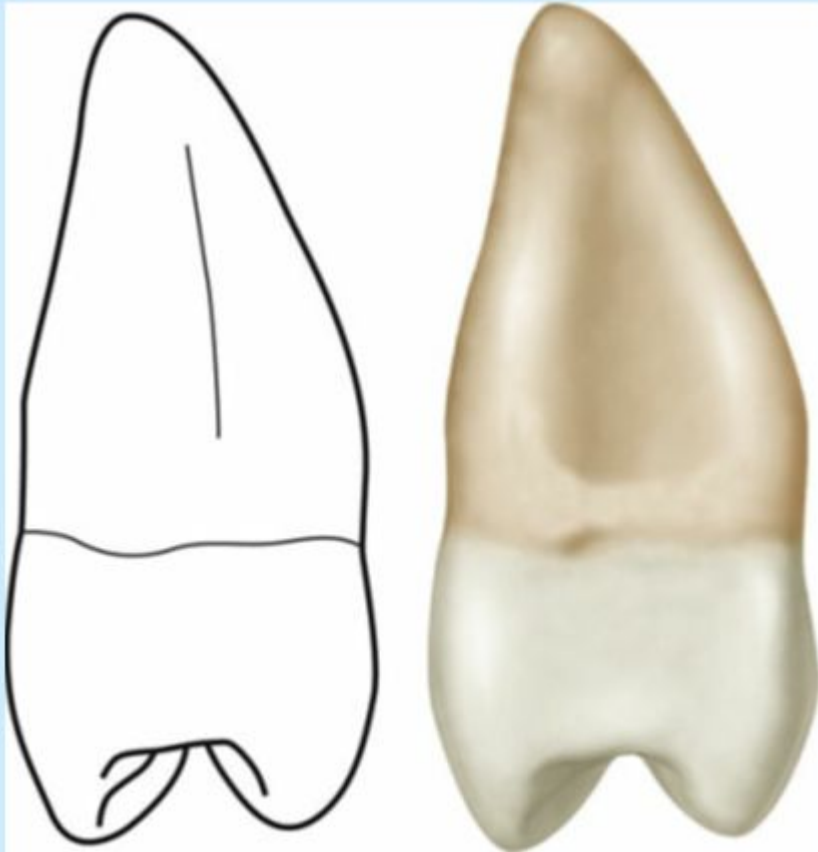
DISTAL ASPECT

- The Distal root surface depression is deeper than the Mesial depression on the maxillary 2nd PM.
- This characteristic of the Distal root surface is the opposite of that of the maxillary 1st PM.
- **This feature has clinical importance during scaling and root planning. As Knowledge of root depressions helps periodontal instrumentation.**

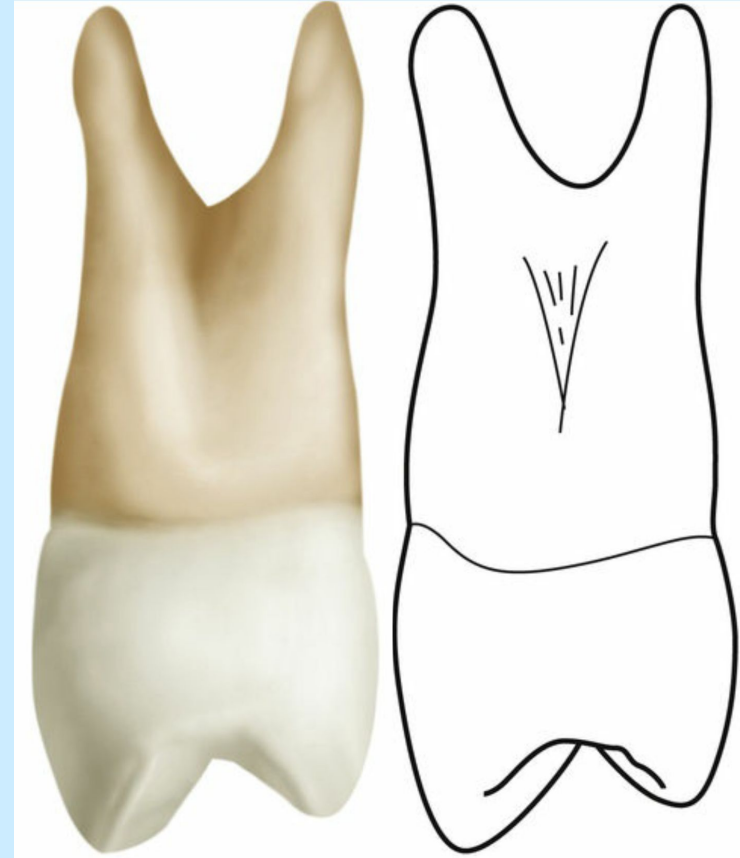


Distal Aspect

Max 2nd PM



Max 1st PM

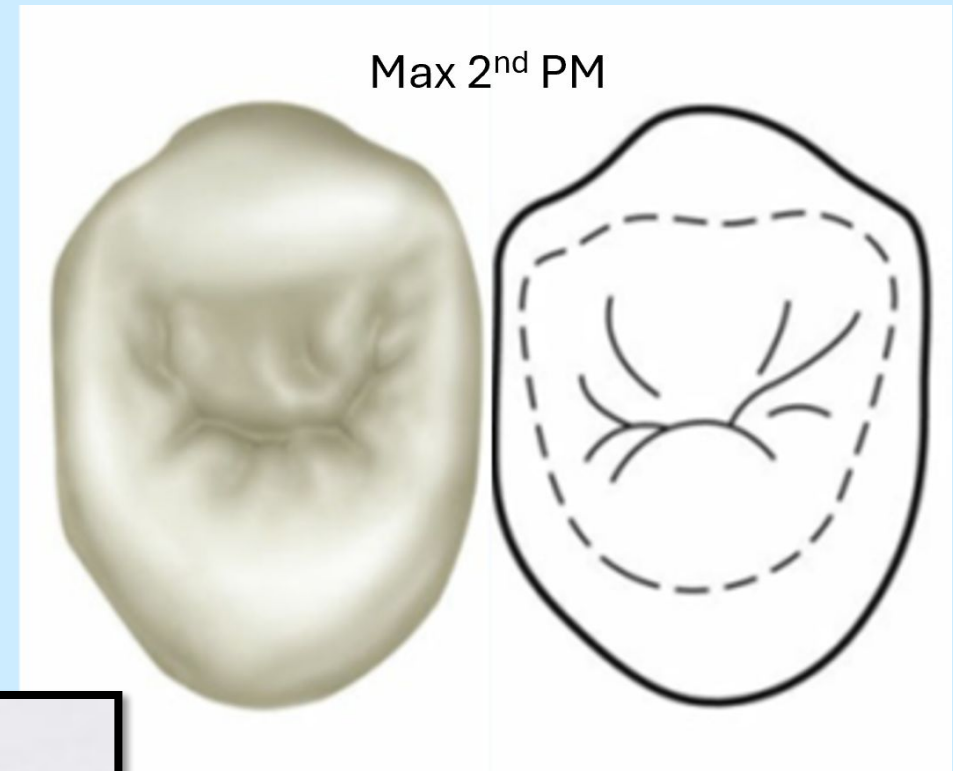
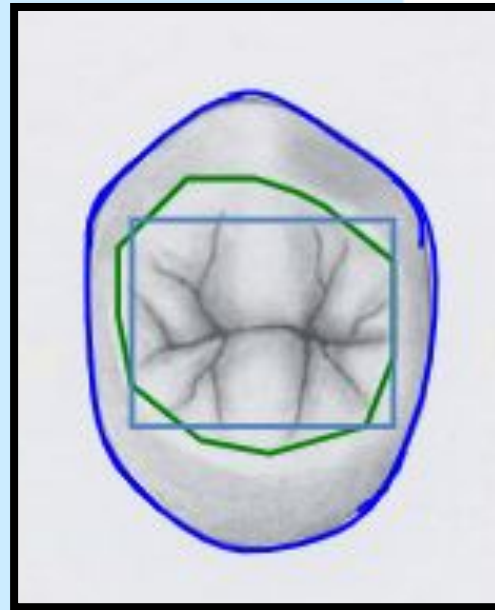


Maxillary left second premolar

OCCLUSAL ASPECT

Occlusal table Outline:

- **rectangular.**
- “**balanced**” look about the tooth from the occlusal aspect.

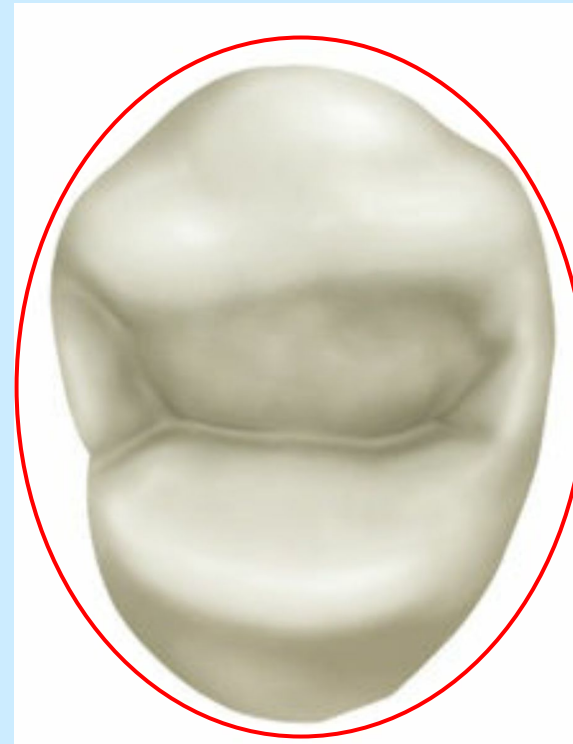


Occlusal Aspect

Outline Less angular, more rounded



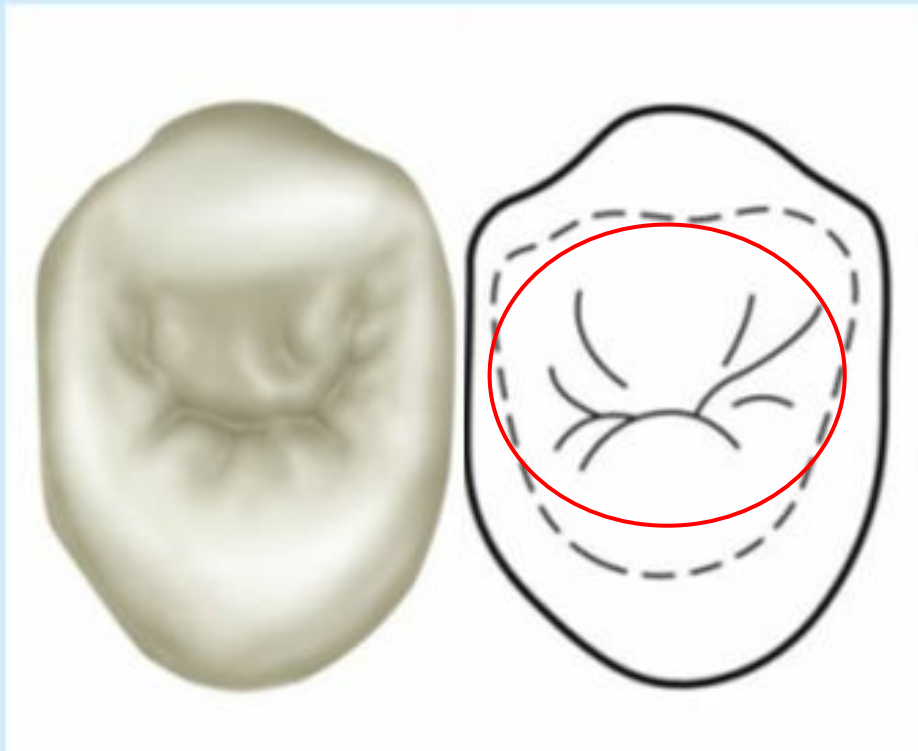
Maxillary left second premolar



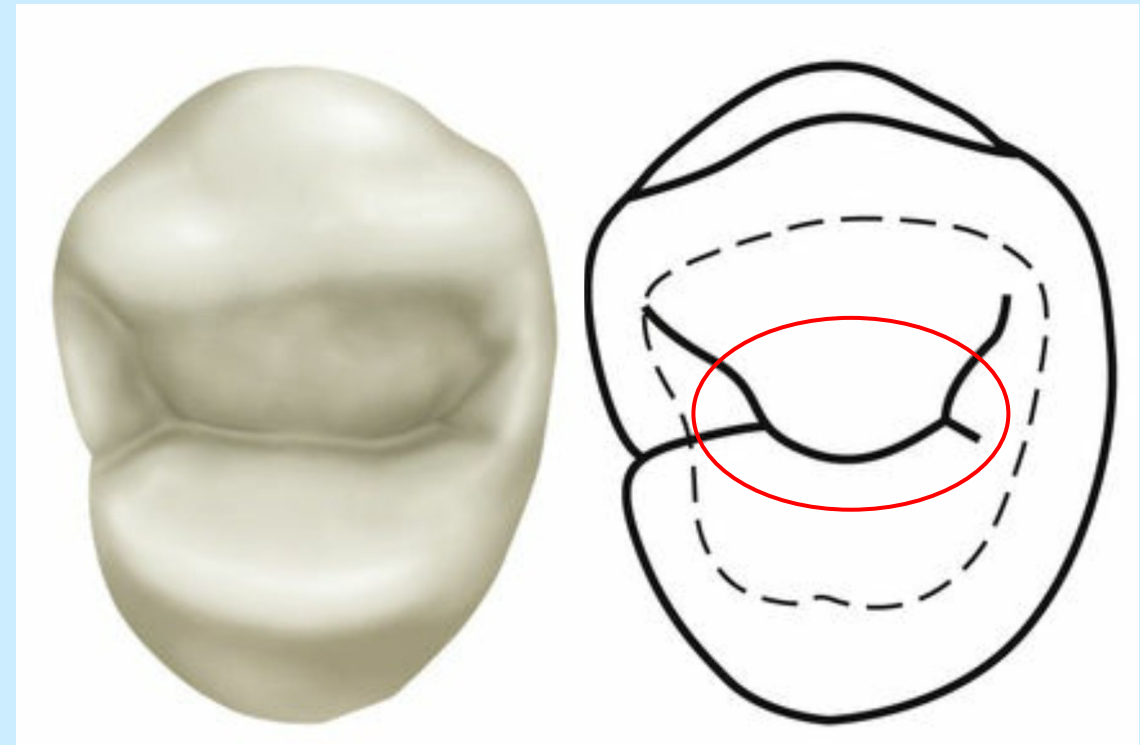
Maxillary left First premolar

Occlusal Aspect

- Occlusal surface irregular with a wrinkled appearance
- Multiple supplemental grooves radiating from central groove



Maxillary left second premolar

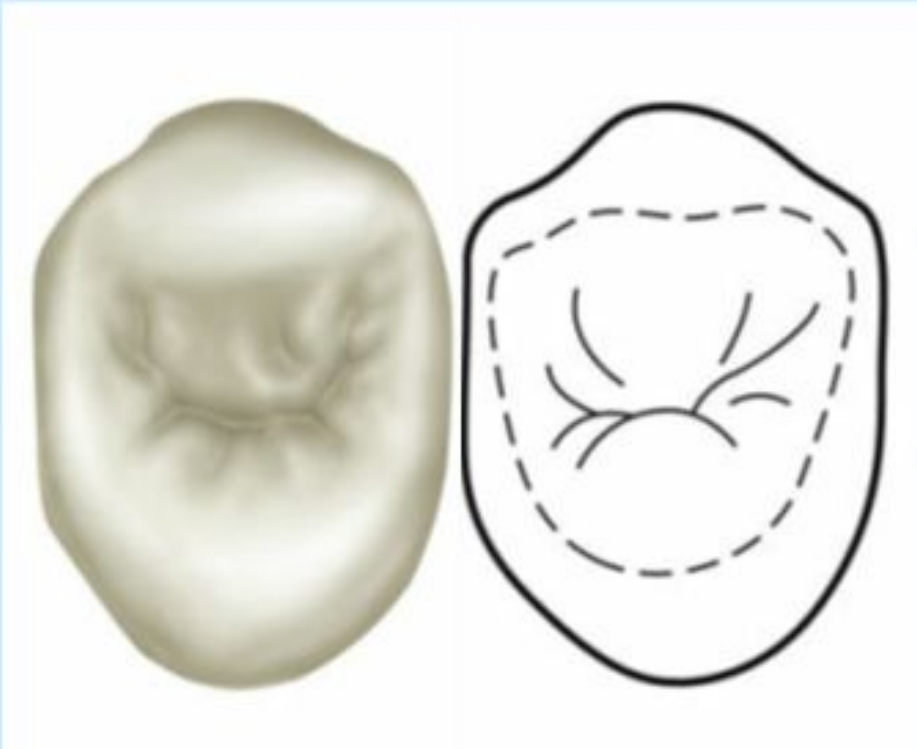


Maxillary left First premolar

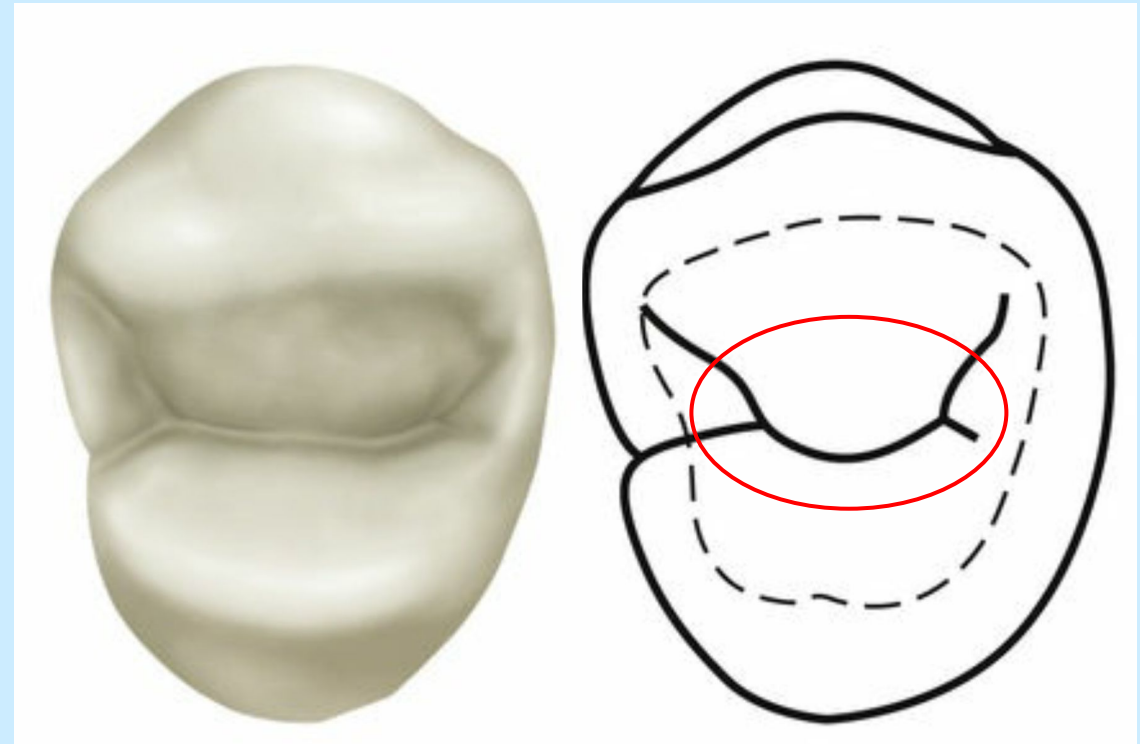
Occlusal Aspect

Mesial & distal triangular fossae:

□ irregular, closer together → central groove shorter

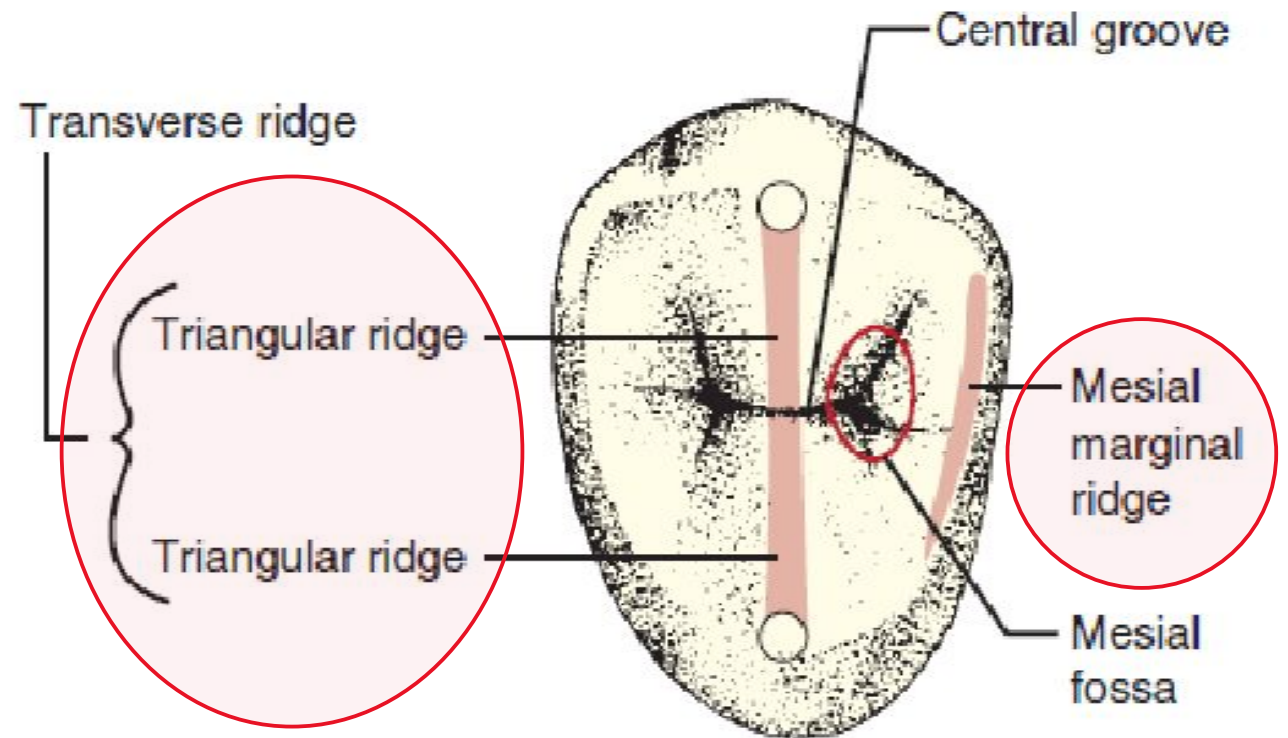


Maxillary left second premolar



Maxillary left First premolar

Triangular ridges of B & L cusps on one side & M Marginal ridge on other side bind the “*Mesial Fossa*”.



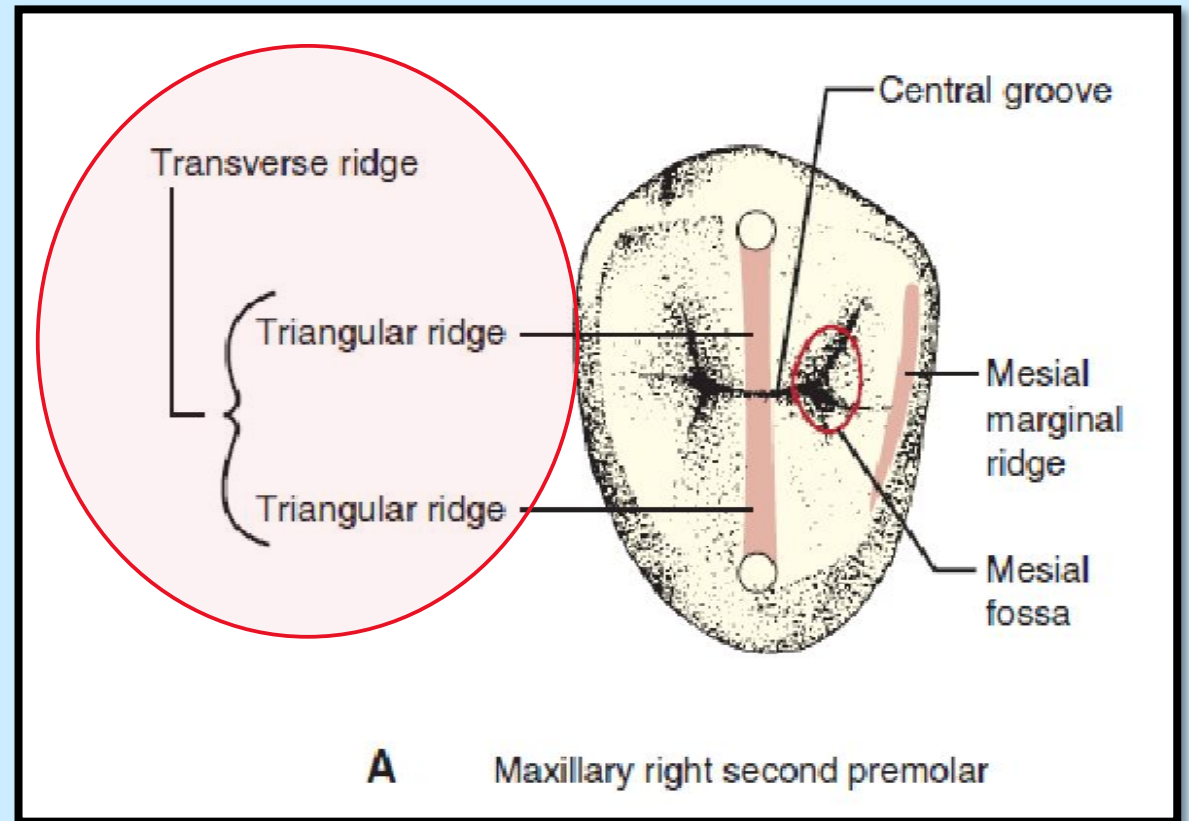
A

Maxillary right second premolar

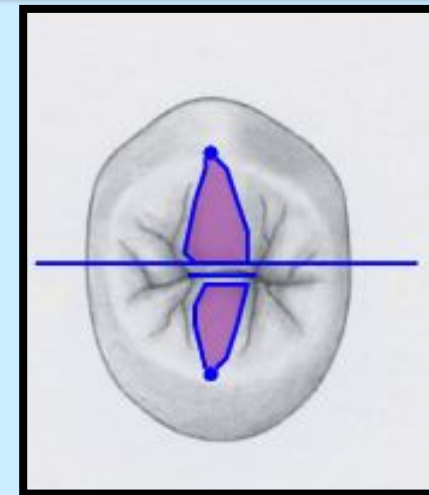
OCCLUSAL ASPECT

Transverse Ridge:

True TR rare as the 2 triangular ridges separated by Central Groove.



B & L Triangular Ridges:
Approx. same length

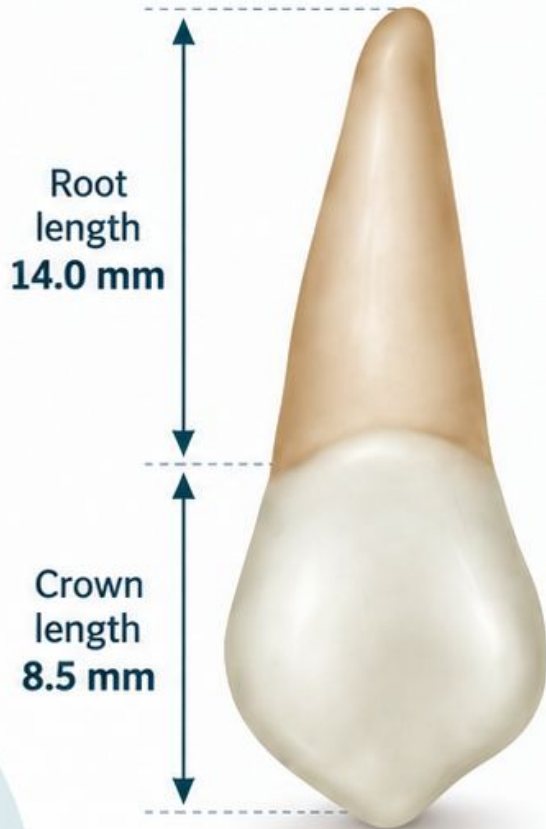


Dimensions and Common Variations

Maxillary Second Premolar | Morphology and Identification | First-Year BDS

Average Dimensions

Maxillary Second Premolar (mm)



Crown length	8.5 mm
Root length	14.0 mm
Mesiodistal crown diameter	7.0 mm
Mesiodistal diameter at cervix	5.0 mm
Buccolingual crown diameter	9.0 mm
Buccolingual diameter at cervix	8.0 mm
Curvature of cervical line, mesial	1.0 mm
Curvature of cervical line, distal	0.0 mm

Common Variations

Examples from Clinical Specimens



Root dwarfing
(short root)



Malformed root



Broad occlusal
surface



Extremely long root



Marked root
curvature



Apical bifurcation



Typical anatomy is important,
but clinical specimens may vary considerably.

Maxillary First vs Maxillary Second Premolar

Key Differences at a Glance

ANTERIOR TO
MOLAR FUNCTION

A key member
of the premolar
family

Feature	First Premolar	Second Premolar
Crown outline	More angular	More rounded
Buccal cusp	Sharper and longer	Shorter and less pointed
Buccal and lingual cusp height	Greater difference	More nearly equal
Central groove	Longer	Shorter
Supplementary grooves	Fewer	More numerous
Occlusal surface	Less wrinkled	More wrinkled
Mesial marginal groove	Commonly present	Usually absent
Mesial crown developmental depression	More evident	Usually absent
Roots	Commonly bifurcated or two roots	Usually one root



Key Takeaway

The maxillary second premolar is **rounder, smoother externally, and more wrinkled occlusally.**

Form
Guides
Function

Maxillary
First
Premolar



Maxillary
Second
Premolar

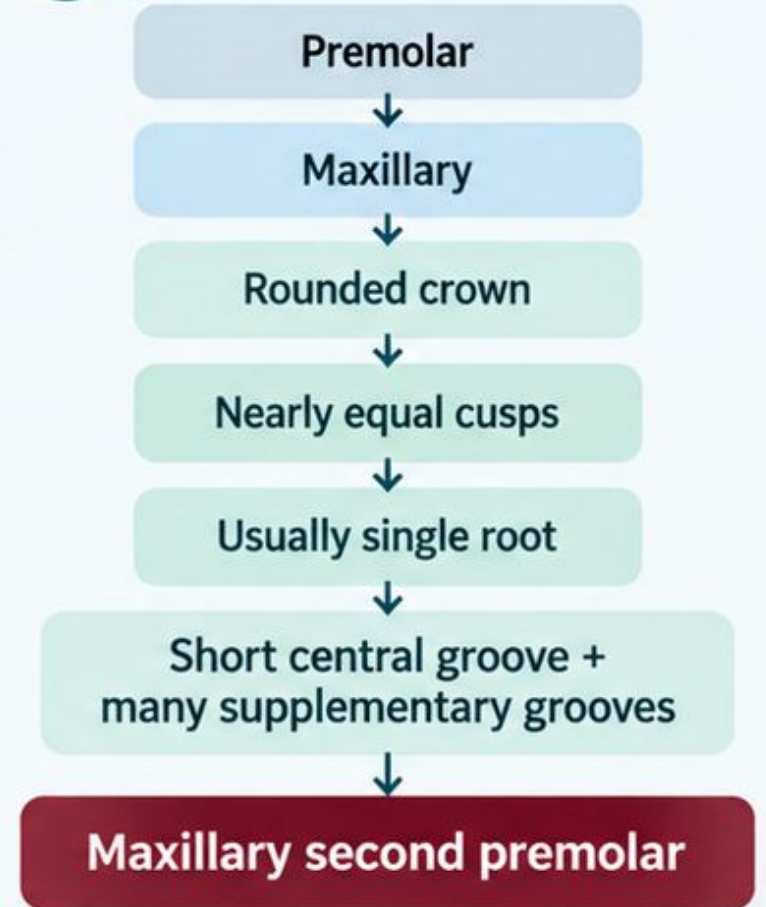


Key Takeaway:

1 5 features you must remember

- 1 Rounded crown
- 2 Usually one root
- 3 Buccal and lingual cusps nearly equal
- 4 Short central developmental groove
- 5 Numerous supplementary grooves giving a wrinkled occlusal appearance

2 Identification pathway



Rounder outside, wrinkled inside.

Scenario

While preparing a restoration, a dentist observes that a bicuspid tooth has an **occlusal surface with multiple supplemental grooves**

Question: Which morphological feature is being described?

Options:

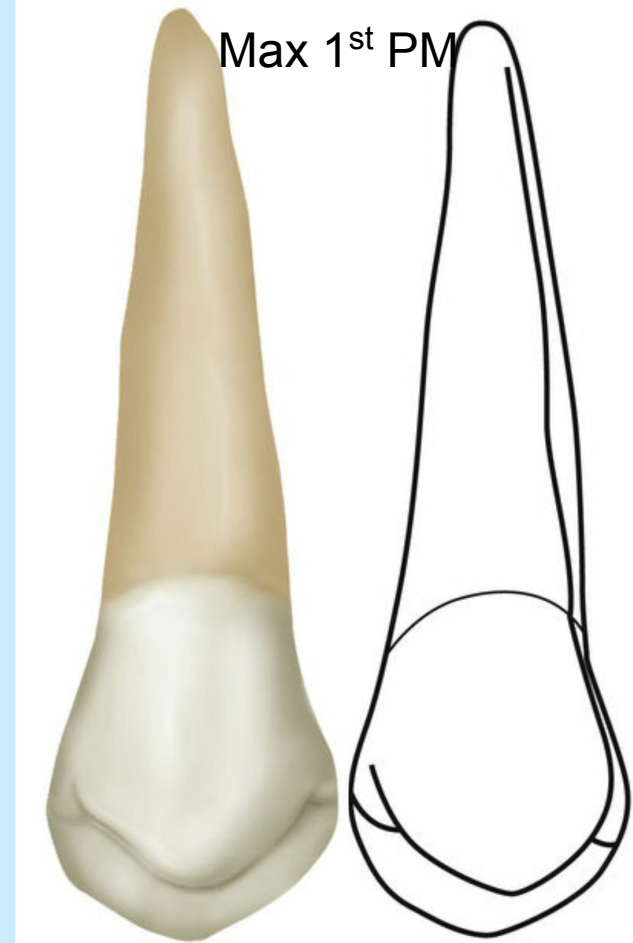
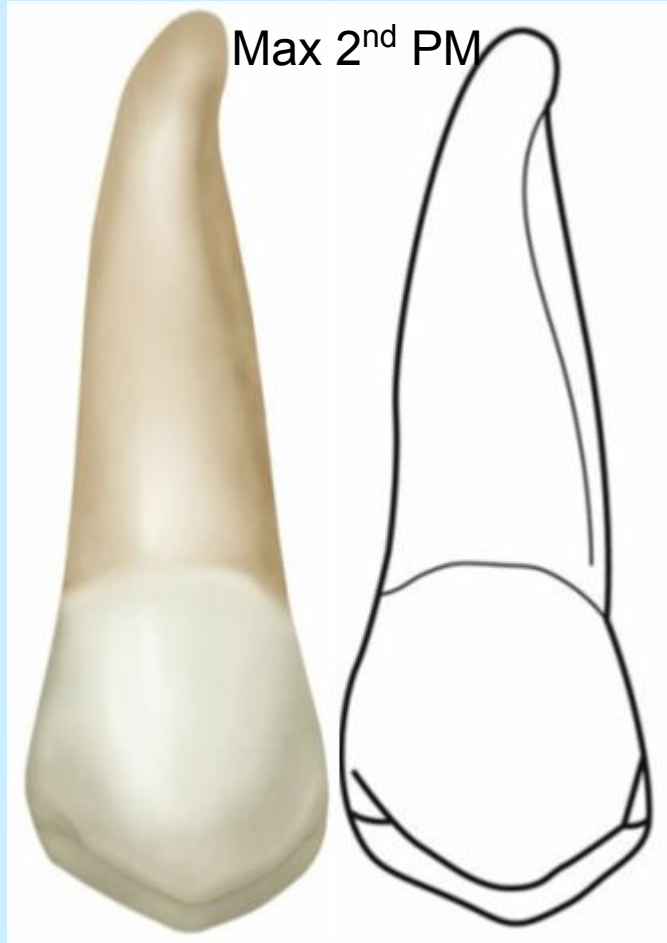
- A. Prominent Transverse ridge with adjacent developmental grooves
- B. Wrinkled occlusal surface of maxillary 2nd premolar
- C. Triangular fossae of maxillary 2nd premolar
- D. Mesial marginal groove of maxillary 1st premolar
- E. Triangular ridges of mandibular premolar



Viva questions

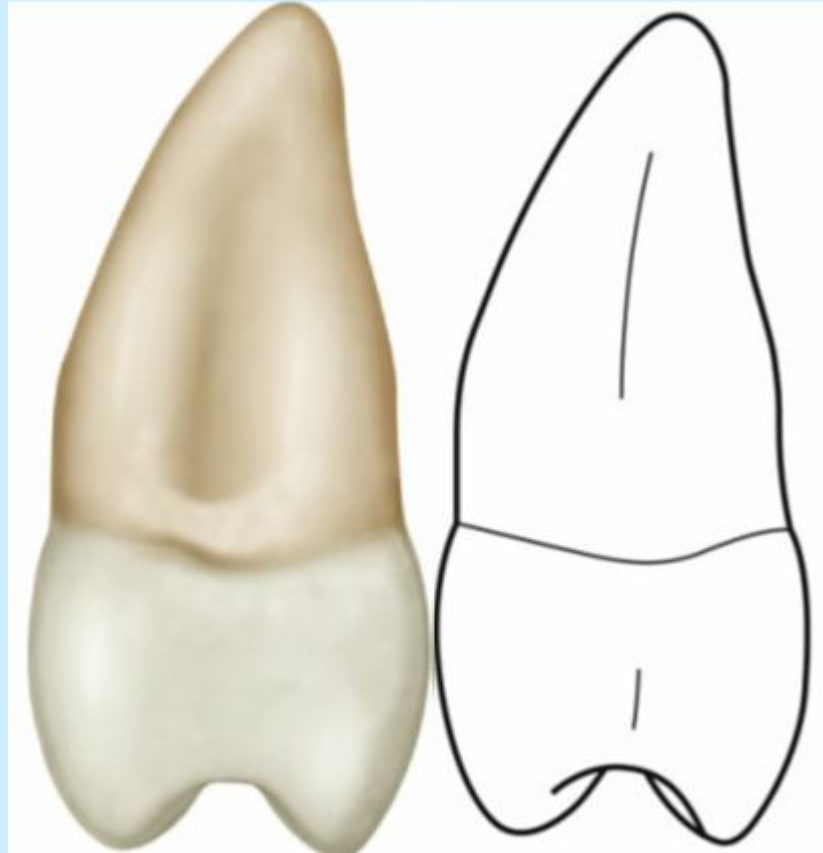
- 1 How many roots are usually present?
- 2 What is the key occlusal identifying feature?
- 3 How does it differ from the maxillary first premolar?

Lingual Aspect

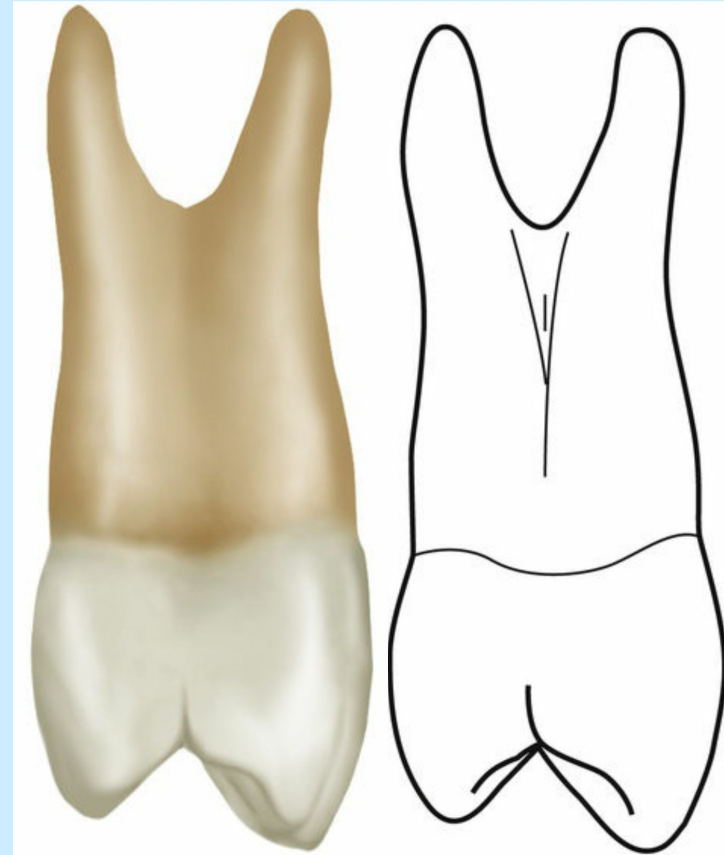


Mesial Aspect

Max 2nd PM

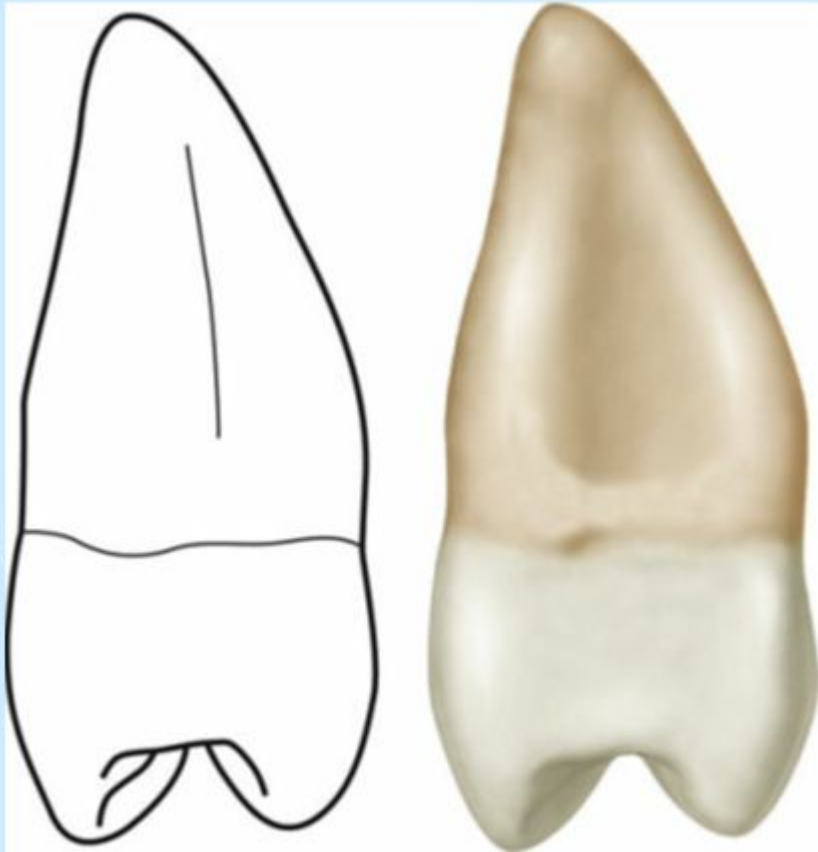


Max 1st PM

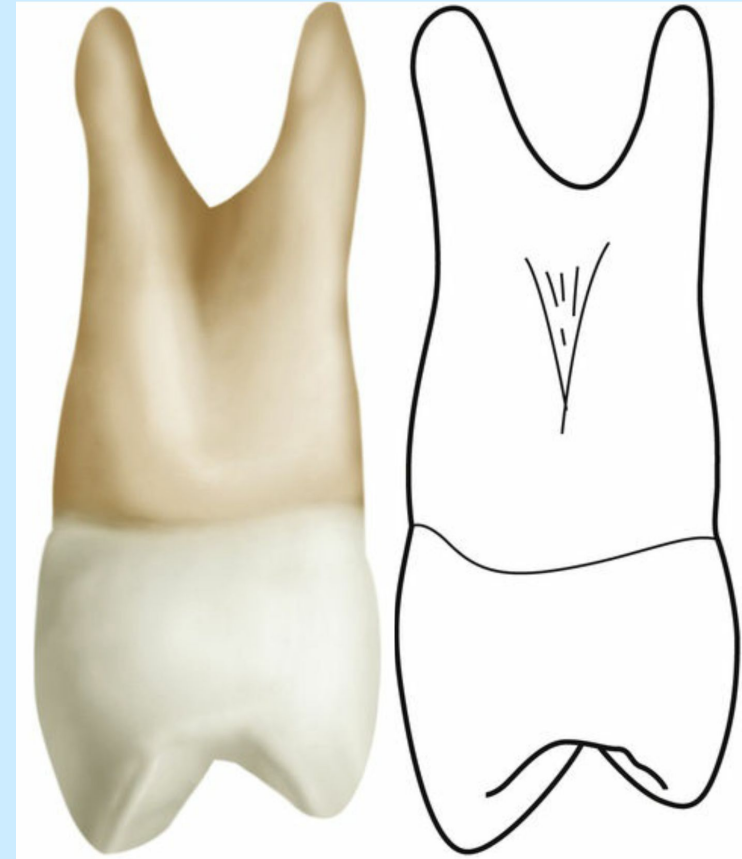


Distal Aspect

Max 2nd PM

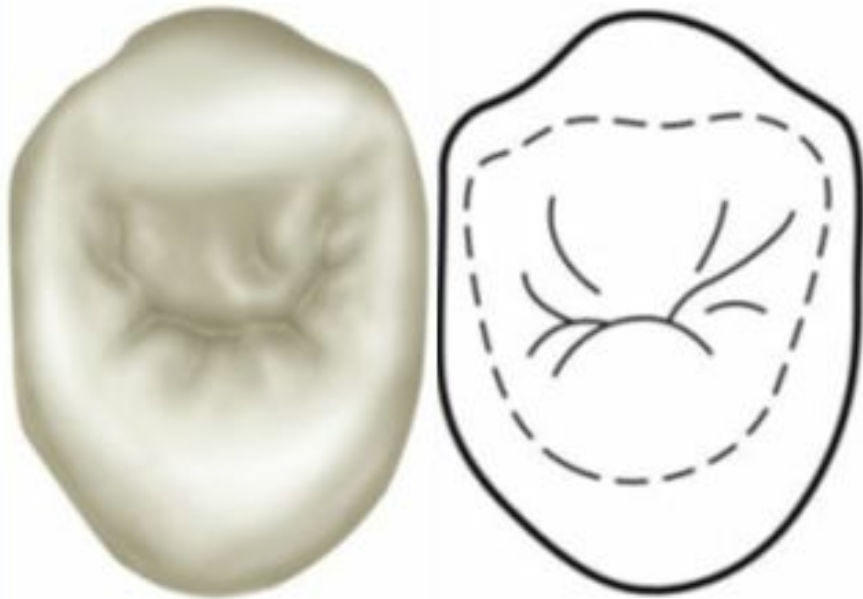


Max 1st PM



Occlusal Aspect

Max 2nd PM



Max 1st PM

